GGG-W-636E
January 13, 1975
SUPERSEDING
Fed. Spec. GGG-W-636D
March 26, 1963

FEDERAL SPECIFICATION

WRENCHES (BOX, OPEN END, AND COMBINATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nonadjustable wrenches used for tightening and loosening bolts and nuts.

1.1.1 Federal specification coverage. Federal specifications do not include all types and sizes of the commodity indicated by the titles of the specifications, nor all those which are commercially available, but includes only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Nonadjustable wrenches shall be of the following types, classes, and styles as specified (see 6.1):

Type I - Box wrench, double head.
  Class 1 - 15° offset each end.
    Style 1 - Regular length.
    Style 2 - Short length.
  Class 2 - Double offset each end, deep or modified offset.
    Style 1 - Regular length.
    Style 2 - Short length.

Type II - Box wrench, double head, 12-point, half moon.

Type III - Combination wrench, open end and 15° offset box opening.

Type IV - Engineers' wrench, double head, open end, 15°.

Type V - Engineers' wrench, single head, open end, 15°.

Type VI - Ignition wrench, double head, open end, 15° and 60° heads.

Type VII - Setscrew wrench, double head, open end, 22-1/2°.

Type VIII - Setscrew wrench, single head, 22-1/2°.
  Class 1 - Open end.
  Class 2 - Box.

Type IX - Structural wrench, offset handle.
  Class 1 - Open end, single head.
  Class 2 - Box, 12-point, single head.

Type X - Tappet wrench, open end, 15°.
  Class 1 - Double spearhead, open end, 15°.
  Class 2 - Single spearhead, open end, 15°.

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Type XI - Slugging wrench, box 12-point.
Style 1 - Short length.
Style 2 - Long length.

Type XII - Striking wrench, box 12-point.

Type XIII - Packing-nut wrench, single head, open end, modified, 12-point, 30° angle.

Type XIV - Flare-nut wrench, single head, open end, modified, 12-point.

Type XV - Thin double head wrench, open end, 3/16-inch offset, 15° angle.

Type XVI - Angle wrench, double head, open end.
Style 1 - 15° and 60°.
Style 2 - 30° and 60°.

Type XVII - Thin head wrench, open end.
Class 1 - Double head, 15° heads (jam or chuck nut).
Class 2 - Single head, straight.
Class 3 - Single head, 30° head.
Class 4 - Single head, 15° head (jam or chuck nut).

Type XVIII - Box wrench, stub end type, single head, 12-point.
Class 1 - Straight handle.
Class 2 - Offset handle.

Type XIX - Handle, extension.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standards:

Fed. Std. No. 346 - Gages, Wrench Opening.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards:

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Unless otherwise specified hereinafter, wrenches shall be steel. The chemical composition of the steel and the heat treatment employed shall be such as to produce wrenches conforming to the physical requirements hereinafter specified.

3.2 Illustrations. The illustrations shown herein are descriptive and not restrictive and are included for the convenience of requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of wrenches which are otherwise in accordance with this specification.

3.3 Design. Wrenches shall be so designed as to afford a well proportioned, comfortable handgrip, and be similar to the figure to which reference is made. A plus or minus tolerance of 5° will be permitted in specified head angle of open end wrenches and offset angle of box wrenches. The nut-and-bolt head-engaging surfaces of the box and open end wrenches shall be finished in a smooth and well-defined manner. The corners and corrations in the box wrench openings shall be clearly defined (not smeared or torn). Wrenches shall withstand the test loads specified in the applicable tables without breaking or permanent deformation (set) which may affect their durability or serviceability when tested as specified in 4.7.2. Wrenches that have a box end design, except type I, class 2, type II, class 2 and types IX, class 2, XI, and XII shall be chamfered on both sides on all points to provide a lead for the working surfaces. The chamfer for type I, class 2 shall be as shown in figure 1. No chamfer is required on types XI and XII. The tips (working ends) of all open end wrenches shall also be chamfered or rounded to eliminate burrs (see figure 1).

3.4 Sizes and wrench-opening tolerances. The sizes of the wrenches shall be as shown for the individual type, style, and class, as applicable. Wrench openings (distance across the flats) shall conform to the dimensions specified herein, and shall be sized to pass the gaging requirements of Fed. Std. 146, except type XV shall conform to the applicable requirements of table XXIV.

3.5 Hardness. Exept as specified for types VI, X, XI, XII, and XVII, classes 3 and 4, and type XIX (see 3.20.1, 3.21, 3.26.3, and 3.26.4), the Rockwell C reading on any portion of the surface of the wrenches shall be not less than 40 nor more than 55 when tested as specified in 4.7.1.

3.5.1 Hardness (powdered metal). Powdered metal tools shall be hardened to not less than 30 nor more than 50 on the Rockwell "C" scale determined by averaging at least 3 tests taken at close proximity to one another.

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3.6 Finish.

3.6.1 Surface finish (see figure 1). All surfaces shall be thoroughly cleaned, free from cracks, and essentially free from burrs, pits, and nodules.

3.6.1.1 Minimum area of surface finish. Minimum areas of surface finish shall be as specified in table I.

3.6.1.2 Forge flash. Forge flash shall be completely removed from the periphery of the heads of all box wrenches, from the circumference of all open end wrenches, and from that portion of the handle which shall be essentially straight and uniform in sectional dimensions. Any remaining forge flash on any surface shall blend smoothly with adjacent surfaces; external sharp edges shall be broken to 1/64 inch radius minimum, and shall not project more than 1/64 inch from adjacent surfaces.

3.6.2 Coatings. The coatings shall be adherent, smooth, continuous, and free from uncoated areas, pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability. The minimum thickness of the coating shall be as specified in 3.6.2.1 through 3.6.2.4 on all external surfaces which can be touched up by a ball 0.750 inch in diameter.

3.6.2.1 Bright chromium plate. The coating shall be electrodeposited metals consisting of nickel, followed by chrome, the minimum thickness being 0.0002 inch for nickel, and 0.00001 inch for chrome; nickel iron undercoating may be substituted.

3.6.2.2 Zinc plate. The coating shall be electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.6.2.3 Oxide coating. The coating shall consist of a chemically produced oxide followed with a coating of preservative oil.

3.6.2.4 Paint. Paint shall be good quality and suitable for adhering to metal surfaces. The coating shall not be tacky nor flake off easily. The coating may be paint, enamel, or a synthetic variety of equivalent quality.

3.6.2.5 Wrenches requiring chromium finish. Unless otherwise specified (see 6.1), the following wrenches shall be chromium plated: types I through IV, VI, X, XV, XVI, and XVII, classes 2 and 3. All other types and classes shall have one of the finishes specified in 3.6.2.1 thru 3.6.2.4.

3.7 Marking. Each wrench head, except for type III combination, shall be marked on one of the faces or on the handle as close to each head as is practicable in a plain and permanent manner with the respective wrench opening as shown in the first column of the applicable table. Type III combination wrenches shall be marked on either one end or both ends. In addition to size markings, each wrench shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.8 Sets. When sets of box and open end wrenches are specified (see 6.1), the components of the sets shall conform to the requirements of the respective items covered in this specification.

3.9 Wraps or rolls. Box and open end wrenches when ordered in sets shall be furnished in plastic wraps or rolls, as specified (see 6.1). The wrap or roll shall be designed to contain wrenches in individual pockets, and may be fabricated of vinyl plastic or cloth duck material unless one of the materials is specified for a particular set. The pockets shall be individually stitched or heat-sealed over their full depth, and their size and number shall be governed by the wrenches in each set. The edges of the body and top of the pockets of the rolls shall have a stitched binding. The wrenches must be easily removable and replaceable in the pockets. Each roll shall be furnished with a permanently attached tie strap or buckle and strap for neatly securing the roll with wrenches. The strap shall be cotton, fungus-and-mildew-resistant treated. The body of each wrap or roll shall be permanently and legibly marked with the Federal Stock Number of the applicable wrench set. The body shall be a .018 inch thick (±0.001 inch) plastic, color optional. The pockets shall be a minimum of 0.012 inch thick with a matte clear (40 percent to 50 percent clear) finish. The plastic shall be flame retardent. Stitching shall be a minimum of 4 to 6 stitches per inch with the thread to be fungus-and-mildew resistant. The wrap or roll shall meet the test requirements, as specified in 4.7.3 and 4.7.4.

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3.10 Type I, box wrench, double head. The wrench shall be suitable for use with hexagonal boltheads and nuts. Each box opening shall have 12 points (double hexagon), except that 7/16 inch or smaller wrenches may have either 6- or 12-point (hexagon or double hexagon) box opening.

3.10.1 Class 1, 15° offset each end, styles 1 and 2, regular and short length. Class 1 wrench shall be similar to figure 2 or figure 2a. Style 1 shall conform to table II and style 2 shall conform to table III. In addition, both styles 1 and 2 shall conform to the test loads of table IA.

3.10.2 Class 2, double offset each end. Deep offset or modified offset shall be furnished in accordance with 3.10.2.1 or 3.10.2.2 at the option of the manufacturer.

3.10.2.1 Deep offset, styles 1 and 2. Wrench shall be similar to figure 3. Style 1 wrench shall conform to table IV and style 2 shall conform to table V. In addition, both styles 1 and 2 shall conform to the test loads of table IA.

3.10.2.2 Modified offset, styles 1 and 2. Wrench shall be similar to figure 4. Style 1 wrench shall conform to table VI and style 2 shall conform to table VII. In addition, both styles 1 and 2 shall conform to the test loads of table IA.

3.11 Type II box wrench, double head, 12-point half open. The wrench shall be suitable for hexagonal boltheads and nuts. Each box opening shall have 12 points (double hexagon). The wrench shall conform to table IA and table VIII and shall be similar to figure 5.

3.12 Type III combination wrench, open end, and 15° offset box opening. The wrench shall have one open end and one box end of identical nominal size. The open end shall be machined to the dimensions, and shall be suitable for hexagon and square-headed bolts and nuts. The box opening shall have 12 points (double hexagon), except that 7/16 inch or smaller wrenches may have either 6- or 12-point (hexagon or double hexagon) box opening. The wrench shall conform to table IA and table IX and shall be similar to figure 6.

3.13 Type IV engineers' wrench, double head, open end, 15°. The wrench shall conform to table X, shall be similar to figure 7, and shall be suitable for hexagonal and square-headed bolts and nuts.

3.14 Type V engineers' wrench, single head, open end, 15°. The wrench shall conform to table XI, and shall be similar to figure 8, and shall be suitable for hexagonal and square-headed bolts and nuts. Wrenches with openings 1-1/2 inches and less shall have flared handles. Wrenches with openings larger than 1-1/2 inches shall have tapered handles.

3.15 Type VI ignition wrench, double head, open end. Wrenches shall be 15° and 60° heads, shall conform to table XII, and shall be similar to figure 9. Wrenches shall be suitable for turning hexagonal and square-headed bolts and nuts.

3.16 Type VII setscrew wrench, double head, open end, 22-1/2°. The wrench shall conform to table XIII and shall be similar to figure 10. The wrench shall be suitable for square-headed setscrews.

3.17 Type VIII setscrew wrench, single head, 22-1/2°.

3.17.1 Class 1, open end. The wrench shall conform to table XIV and shall be similar to figure 11. The wrench shall be suitable for square-headed setscrews.

3.17.2 Class 2, box wrench. The wrench shall conform to table XIV and shall be similar to figure 12. Wrench openings shall be 4 or 8-point at the option of the manufacturer.

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3.18 Type IX structural wrench, offset handle. The wrench shall have an offset tapered handle suitable for aligning boltholes.

3.18.1 Class 1, open end, single head. The wrench shall conform to table XV and shall be similar to figure 13. The wrench shall be suitable for hexagonal and square boltheads and nuts.

3.18.2 Class 2, box, single head, 12-point. The wrench shall conform to table XVI and shall be similar to figure 14. The wrench shall be suitable for hexagonal boltheads and nuts.

3.19 Type X tappet wrench, open end, 15°. The wrench shall be suitable for hexagonal and square-headed bolts and nuts. Hardness shall be not less than 45 nor more than 58 on the Rockwell "C" scale.

3.19.1 Class 1 tappet wrench, double spearhead, open end, 15°. The wrench shall conform to table XVII and shall be similar to figure 15.

3.19.2 Type X, class 2 tappet wrench, single spearhead, open end, 15°. The wrench shall conform to table XVIII and shall be similar to figure 16.

3.20 Type XI slugging wrench, 12-point box. The wrench shall be suitable for hexagonal boltheads and nuts. One end shall have a box opening having 12 points (double hexagon) and the other end shall terminate with a surface suitable for slugging. The longitudinal centerline of the handle shall be parallel to the plane of the wrench heads. The wrenches shall be capable of withstanding the impact test specified in 4.7.2.3.2.

3.20.1 The Rockwell C number of any portion of the wrench shall be not less than 28 nor more than 38.

3.20.2 Style 1, short length. Style 1 wrench shall conform to table XIX and shall be similar to figure 17.

3.20.3 Style 2, long length. Style 2 wrench shall conform to table XX and shall be similar to figure 18.

3.21 Type XII striking wrench, 12-point box. The wrench shall be suitable for use with hexagonal boltheads and nuts. One end shall have a box opening having 12 points (double hexagon) and the other end shall have a striking block having a rectangular cross section. The handle with striking block shall be offset with the longitudinal centerline of the handle at an angle of 5° ±5° from the plane of the wrench head. The nut and bolthead engaging surfaces shall be finished in a smooth and well-defined manner and the corners of the serrations shall be sharp. The Rockwell C number of any portion of the wrench shall be not less than 35 nor more than 45. The wrench shall conform to table XXI and shall be similar to figure 19.

3.22 Type XIII packing-nut wrench, single head, open end, modified 12-point, 30° angle. The wrench shall be suitable for use with hexagonal packing nuts. Each opening shall be based on a modified 12-point design. The wrench shall conform to table XXII and shall be similar to figure 20.

3.23 Type XIV flare-nut wrench, single head, modified 12-point. The wrench shall be suitable for use with hexagonal flare nuts. Each opening shall be based on a modified 12-point design. The wrench shall conform to table XXIII and shall be similar to figure 21.

3.24 Type XV, thin double head, open end, 3/16 inch offset, 15° angle. The wrenches shall be suitable for hexagonal or square headed nuts and bolts, and need not be forged. The wrench shall conform to table XXIV and shall be similar to figure 22.

3.25 Type XVI angle wrench, double head, open end. The wrench shall have identical openings at each end.

3.25.1 Style 1, 15° and 60° heads. The wrench shall conform to table XXV and shall be similar to figure 23.

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3.25.2 Style 2, 30° and 60° heads. The wrench shall conform to table XXVI and shall be similar to figure 24.

3.26 Type XVII thin head wrench, open end. The wrench shall be suitable for use on thin hexagon and square head bolts and nuts.

3.26.1 Class 1, double head, 15° (jam nuts). The wrench shall conform to table XXVII and shall be similar to figure 25.

3.26.2 Class 2, single head, straight. The wrench shall conform to table XXVIII and shall be similar to figure 26.

3.26.3 Class 3, single head 30°. The wrench shall have a hardness on any portion of the wrench not less than 44 nor more than 51 on a Rockwell C scale. The wrench shall conform to table XXIX and shall be similar to figure 27.

3.26.4 Class 4, single head 15° (jam nut). The wrench shall have a hardness on any portion of the wrench not less than 39 nor more than 49 on a Rockwell C scale. The wrench shall conform to table XXX and shall be similar to figure 28.

3.27 Type XVIII box wrench, single head, 12-point, stub end, handle. The wrench shall be suitable for heavy duty nut turning on hexagon nuts. Each box opening shall be 12 points. The stub end of the handle shall be round and suitable for accepting the applicable type XVI handle. The handle shall be provided with a plunger stop to prevent type XVI handle from turning or slipping under normal working conditions.

3.27.1 Class 1, straight. The wrench shall conform to table XXXI and shall be similar to figure 29.

3.27.2 Class 2, offset. The wrench shall conform to table XXXII and shall be similar to figure 30.

3.28 Type XIX, handle, extension wrench. The handle shall be of steel tubing designed for use on type XV wrenches and to fit and engage the locking plunger on the wrenches. The handle shall conform to table XXXIII for the applicable size wrench specified (see 6.1), and shall be similar to figure 31.

3.29 Workmanship. The requirements within this specification are intended to describe the best commercial quality wrenches available. The wrenches shall conform to the quality of end product specified by the requirements in this specification. In addition, the wrenches shall be free from rust, burrs, fins, nodules, or other defects which may impair their serviceability, durability, or appearance. The edges and corners of the wrenches shall be clearly defined and smoothly finished with no evidence of broaching tears on the working surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

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4.2 Sampling for lot acceptance inspection.

4.2.1 Lot. All wrenches offered for delivery at one time shall be considered a lot for the purpose of acceptance, inspections, and tests.

4.2.2 Sampling for inspection and tests. A random sample shall be selected from each lot in accordance with the provisions of MIL-STD-105. Sampling shall be as stated in table XXXIV, defects are listed in table XXXV.

4.3 Lot acceptance inspection. Lot acceptance inspection shall be applied to each item in accordance with 4.4.2 and 4.4.3 prior to being offered for acceptance under the contract. Lot acceptance inspection shall consist of the following:

(a) Examination (see 4.4).
(b) Tests (see 4.5).
(c) Inspection of preparation for delivery (see 4.6).

4.4 Examination and testing.

4.4.1 Examination and testing of components.

4.4.1.1 Visual and dimensional examination. An inspection lot shall consist of all like components submitted for inspection at one time. Each sample unit shall be examined for any nonconformance in design, material, dimension, finish, coating, construction, marking, and workmanship. (Examination of decimal dimensions with three or more places shall be performed as outlined in 4.4.1.2.)

4.4.1.2 Testing. Each sample unit shall be tested to determine conformance with specified requirements. When specific test procedures are shown herein, these procedures shall be followed. When specific test procedures are not cited, all such procedures used shall have the approval of the Government.

4.4.2 Examination and testing of sets. An inspection lot shall consist of all sets submitted for inspection at one time. Examination and testing shall be accomplished by considering the inspection lot as being divided into sub-lots (for example, a lot of 300 sets each containing 19 components and one box shall be examined as 20 sub-lots of 300 each.) A sub-lot shall consist of all like components, such as, like tools of a specific type, class, and size. Each sub-lot shall be inspected as individual components, and 4.3 and 4.4.1 will apply. Rejection of any sub-lot, based on the specified Acceptable Quality Level (AQL), shall constitute rejection of the entire lot.

4.5 Test procedures.

4.5.1 Hardness. Wrenches for which hardness requirements are specified shall be tested on a Rockwell tester using the procedures outlined in ASTM E18 unless otherwise specified (3.5). If necessary, plating, decarburization, and hardened cases shall be removed before testing. When grinding to prepare the test surface, the thickness of material removed shall be 0.007 inch maximum on all machined or polished surfaces and 0.016 inch maximum on all other surfaces.

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4.5.2 Loads. Sample wrenches shall be tested for conformance with 3.3 by applying the bending moment test load specified in the applicable table for each type, style, class, and size of wrench.

4.5.2.1 Reference lines for determining permanent deformation. In order to prepare the sample for test, suitable reference lines may be scribed on the jaws and handles. After application of test load, examination for permanent deformation shall be made.

4.5.2.2 Mandrels for wrench openings. Suitable mandrels shall be used to fit into the wrench opening and to provide the proper support and necessary strength for the test load applied. The wrenches shall be tested on hexagonal mandrels, except that types VII and VIII, classes 1 and 2 wrenches shall be tested on square mandrels. Mandrels shall conform to the dimensions and tolerances of table XXVI and XXVII. Mandrels shall be hardened to show a Rockwell hardness of not less than C 55 and smoothly finished on the wrench engaging surfaces.

4.5.2.3 Application of test load.

4.5.2.3.1 For wrenches, except type XI. The test load specified in the applicable table is the torque applied to the test mandrel which tends to rotate the mandrel about its longitudinal axis. The torque shall be applied by inserting the mandrel through the wrenching surfaces. The force required to produce the torque shall be applied as far from the mandrel as practicable. The torque shall be applied at a rate not to exceed 90° per minute and held for a minimum of 10 seconds. The test load may be computed when the longitudinal axis of the mandrel is horizontal and the force direction is vertical (see figure 32) by multiplying the force magnitude by the shortest distance from the mandrel longitudinal axis to the line of action of the force. The wrench shall be considered to have failed the test if there is visible permanent distortion of the handle; permanent deformation of the box head with respect to the handle in excess of 5°; permanent spreading of the open end jaws, at the tips, in excess of 0.002 inch over sizes 5/32 inch to 1 inch inclusive; 0.003 inch, sizes 1-1/16 inches to 3 inches inclusive; 0.004 inch, and sizes 3-1/8 inches to 6 inches inclusive; the "NO GO" gage sizes of Fed. Std. 346; any cracking or breakage of any portion of the wrench; or any sliding of the wrench around the mandrel.

4.5.2.3.2 Type XI, plugging wrench. The mechanics for performing the impact test on type XI wrenches shall consist essentially of a cylindrical weight, an applicable length of a steel pipe, with an inside diameter slightly larger than the diameter of the weight suitably assembled to guide the weight during the drop impact test and a steel base plate of dimensions approximately 4 inches thick, 12-1/2 inches wide, and 32 inches long for supporting the mandrel and wrenches while under the impact test. Three sample wrenches of the same style and size shall be mounted on a mandrel conforming to 4.5.2.2 with the middle wrench offset not more than 30° with respect to the other wrenches. Shims 1/4 inch thick shall be placed under the heads of the two wrenches on the ends of the mandrel and the assembly clamped at each of these heads to the heavy base plate. The plugging block end of the middle wrench shall be subjected to 100 impact blows. Upon completion of the test, the wrench shall be measured at the handle for permanent deformation and examined for damages. The permanent deformation shall be measured approximately 1-1/4 inches in from the plugging block end by means of a dial indicator mounted at a fixed point on the top flat of the hexagon mandrel by light clamping or holding firm with the hand. The wrenches shall withstand 100 impact blows of the magnitude specified in table XXVII for the respective style and sizes without breakage or damage which may affect the serviceability of the wrench and without exceeding 1/16 inch permanent deformation measured at the handles. Either style 1 (short) or style 2 (long) length plugging wrenches will be accepted if they conform to the impact test in table XXVII for the style 2 (long length) wrenches.

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4.5.3 Tensile pull test for wraps or rolls. In addition to the test required to verify compliance for materials and fabrication, at least two pairs of pockets shall be given a test as follows:

A test mandrel consisting of two legs at least 3 inches long and 1/4 to 1/2 inches wide shall be inserted into a pair of pockets on each side of the stitch or seal. The mandrel shall be so formed that a straight tensile pull can be applied, through a spring scale or other suitable measuring device, perpendicular to the pockets. The wrap or roll being tested shall be clamped or firmly secured at each of the four corners and then a straight tensile pull of at least 25 pounds, as registered on the scale, shall be applied. Any evidence of tearing of the pockets, or separation of the stitch or seal, shall be cause for rejection.

4.5.4 Plastic wrap or roll test. Vinyl plastic wrap or roll shall meet the following minimum requirements when tested as follows:

-20	Masland cold crack
1.30 maximum	Specific gravity
±0.001	Gage consistency
1.58 percent maximum	Volatility

4.6 Inspection of preparation for delivery. Inspection shall be made to determine compliance with the requirements of PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size of tool.
(c) Type of coating required (see 3.6).
(d) The number of parts comprising the sets and the size of the wrenches, when sets are required (see 3.8).
(e) If wraps or rolls are not specified for sets (see 3.9).
(f) Level of preservation, packaging, and packing required (see section 5).

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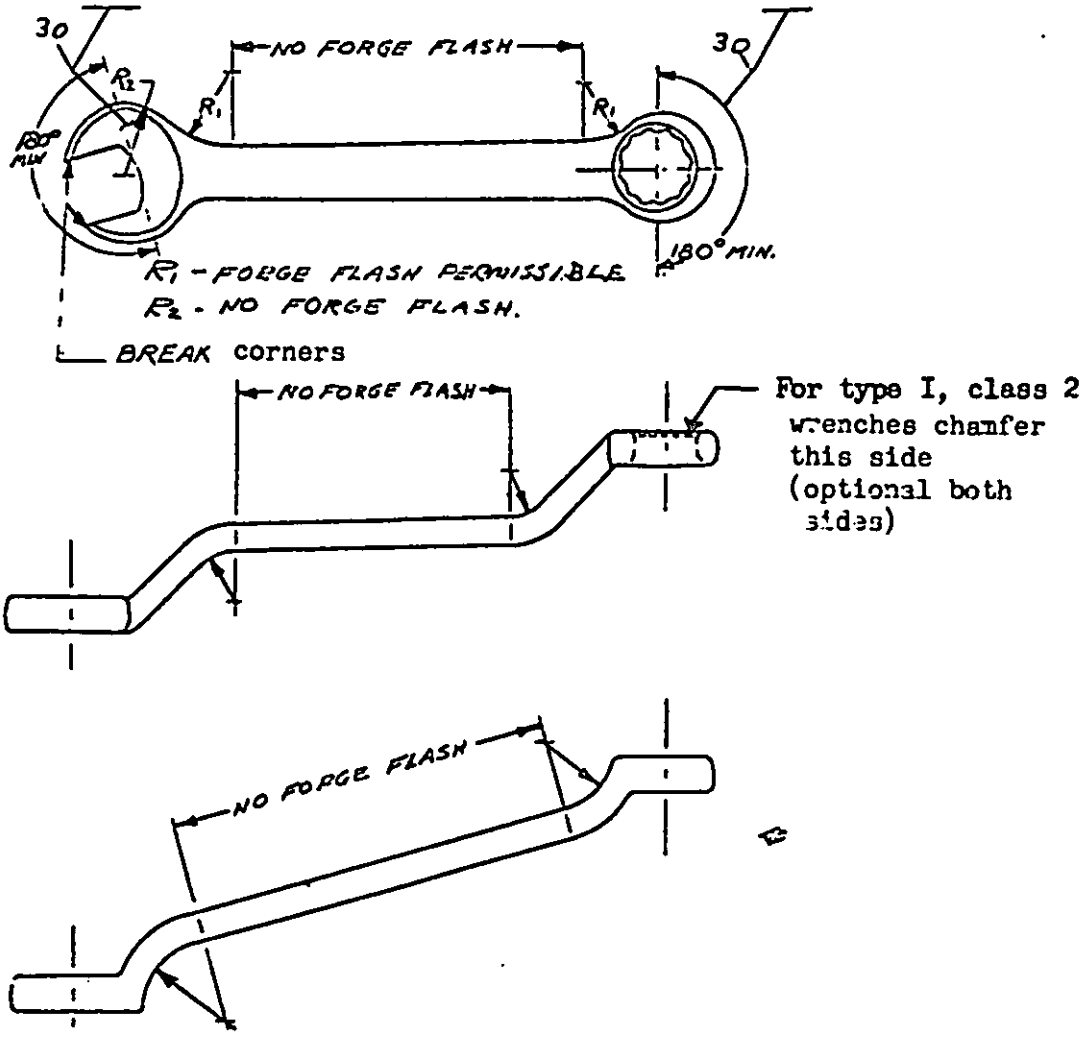


FIGURE 1. Finish requirements.

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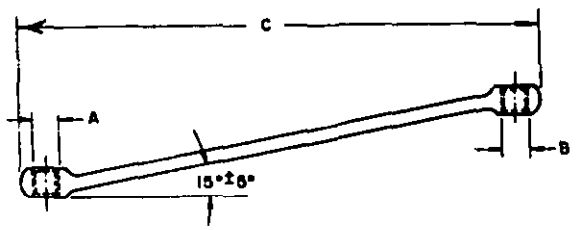


FIGURE 2. Type I, class 1, styles 1 and 2 box wrench, double head, 15° offset each end, regular and short lengths.

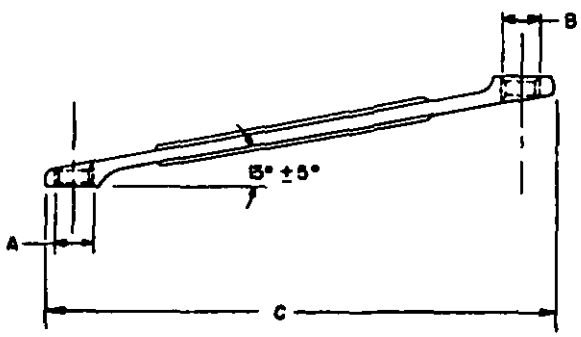


FIGURE 2a. Type I, class 1, styles 1 and 2 box wrench, double head, 15° offset each end, regular and short lengths.

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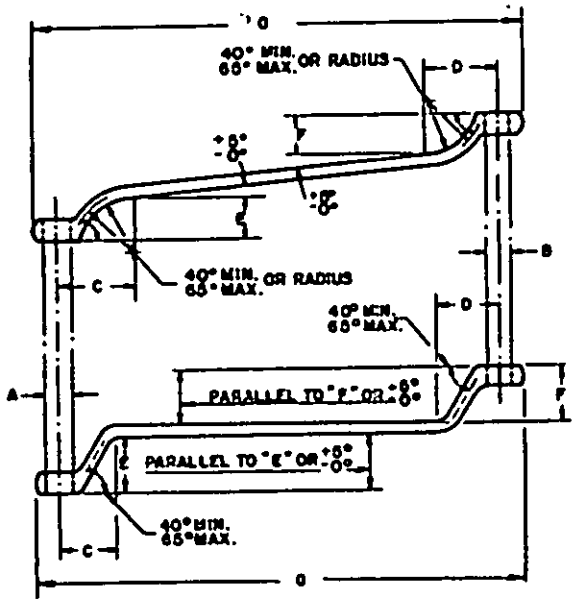


FIGURE 3. Type I, class 2, styles 1 and 2 box wrench, double end, deep offset each end, regular and short lengths.

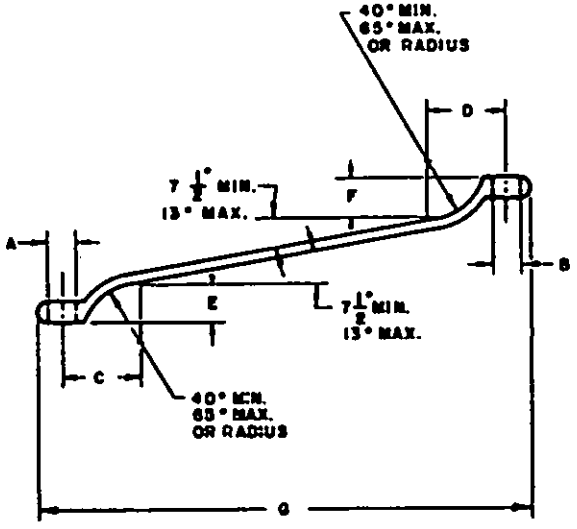


FIGURE 4. Type I, class 2, styles 1 and 2 box wrench, modified offset, regular and short lengths.

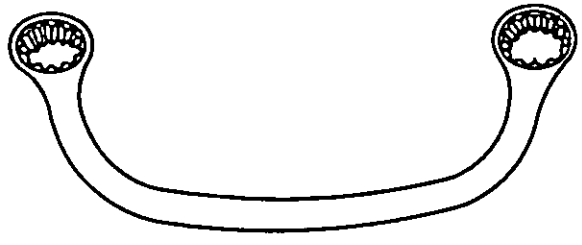


FIGURE 5. Type II box wrench, double head, 12-point, half moon.

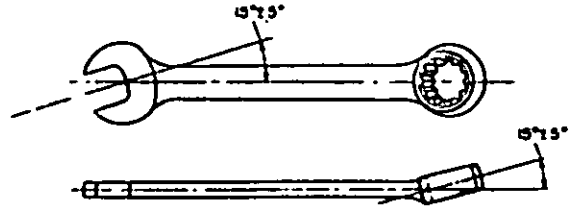


FIGURE 6. Type III combination wrench, open end, and 15° offset box opening.

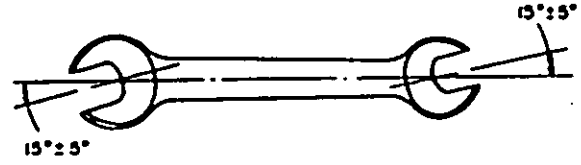
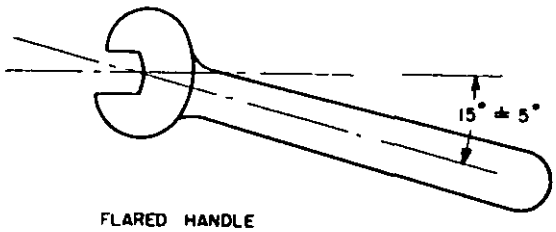
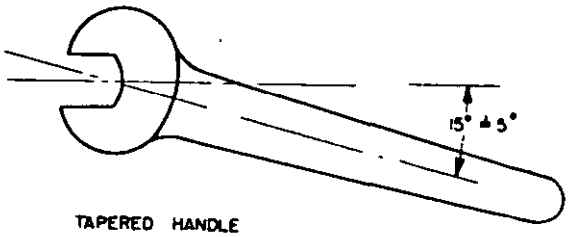


FIGURE 7. Type IV engineers' wrench, double head, open end, 15°.

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FLARED HANDLE



TAPERED HANDLE

FIGURE 8. Type V engineers' wrench, single head, open end, 15°.



FIGURE 9. Type VI, ignition wrench, double spearhead, open end, 15° and 60° heads.

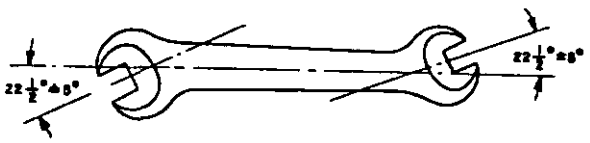


FIGURE 10. Type VII setscrew wrench, double head, open end, 22-1/2°.

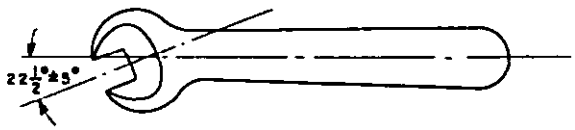


FIGURE 11. Type VIII, class 1 setscrew wrench, single head, open end, 22-1/2°.

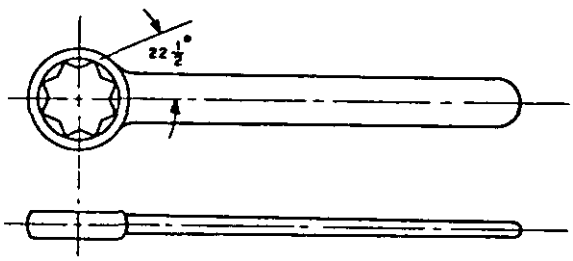


FIGURE 12. Type VIII, class 2 setscrew wrench, single head, box, 22-1/2°, 8-point.

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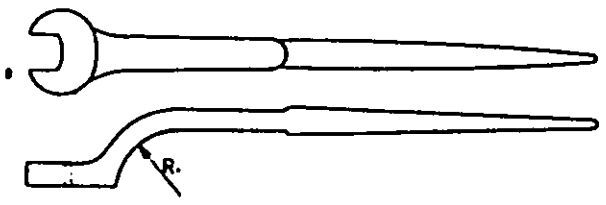


FIGURE 13. Type IX, class 1 structural wrench, offset handle, single head, open end.

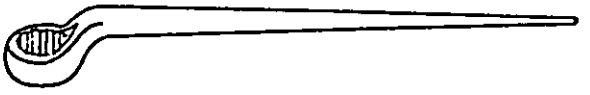


FIGURE 14. Type IX, class 2 structural wrench, offset handle, box, single head, 12-point.

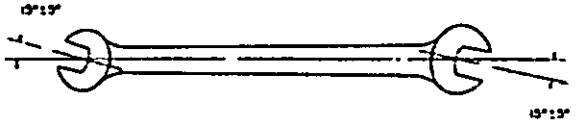


FIGURE 15. Type X, class 1 tappet wrench, double spear-head, open end, 15°.

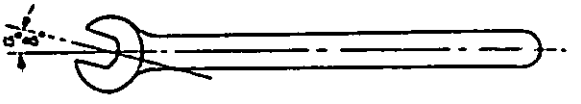


FIGURE 16. Type X, class 2 tappet wrench, single spear-head, open end, 15°.

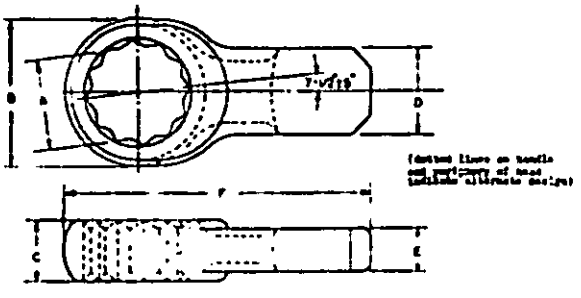


FIGURE 17. Type XI, style 1 slugging wrench, 12-point box, short length.

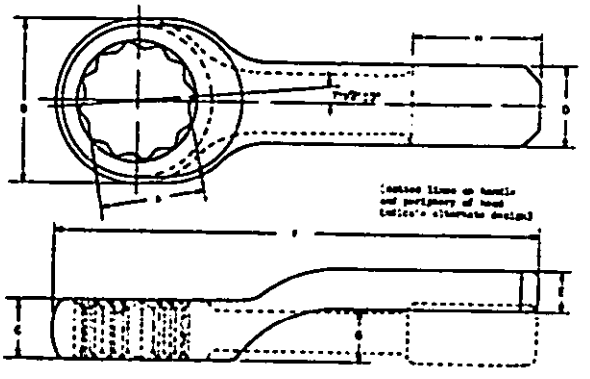


FIGURE 18. Type XI, style 2 slugging wrench, 12-point box, long length.

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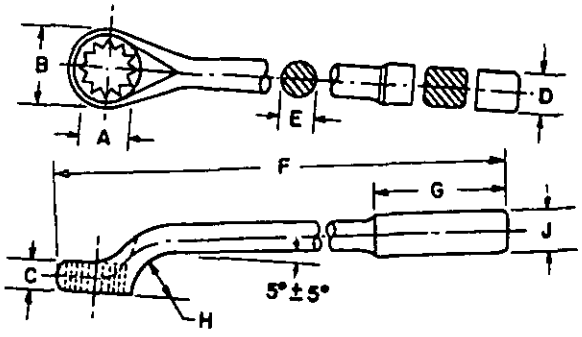


FIGURE 19. Type XII striking wrench, 12-point box.

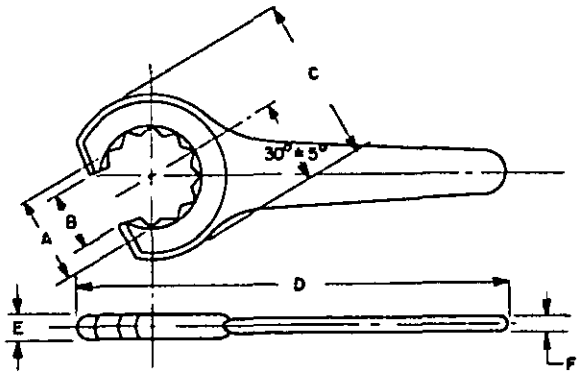


FIGURE 20. Type XIII packing-nut wrench, single head, open end, modified 12-point, 30° angle.

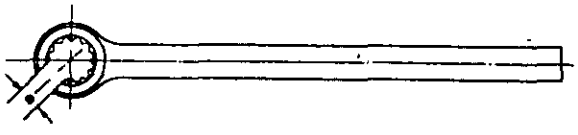


FIGURE 21. Type XIV flare-nut wrench, single head, modified 12-point.

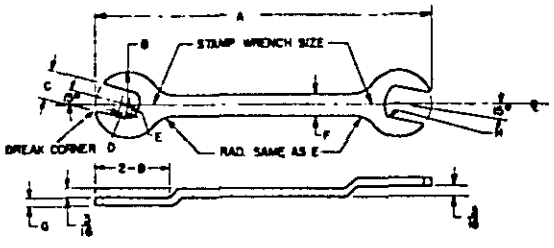


FIGURE 22. Type XV, thin double head, open end, 3/16 inch offset, 15° angle wrench.

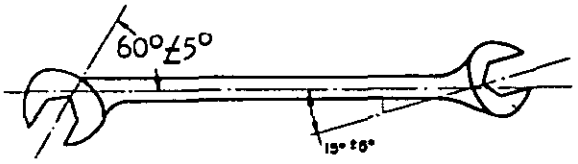


FIGURE 23. Type XVI, style 1 angle wrench, double head, open end, 15° and 60° heads.

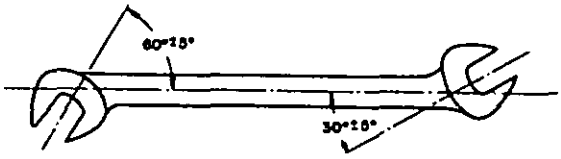


FIGURE 24. Type XVI, style 2 angle wrench, double head, open end, 30° and 60° heads.

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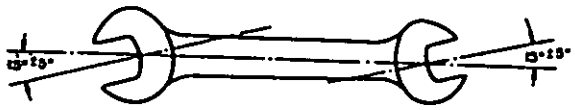


FIGURE 25. Type XVII, class 1 thin head wrench, double head, open end, 15° (jam nut).

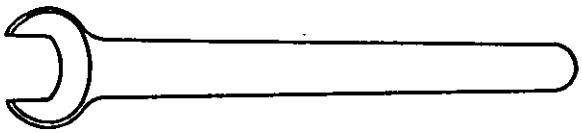


FIGURE 26. Type XVII, class 2 thin head wrench, single head, open end, straight.

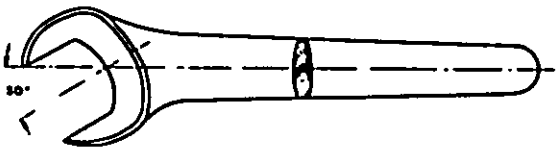


FIGURE 27. Type XVII, class 3 thin head wrench, single head, open end, 30° head.

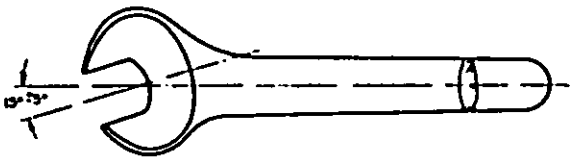


FIGURE 28. Type XVII, class 4 thin head wrench, single head, open end, 15° head.

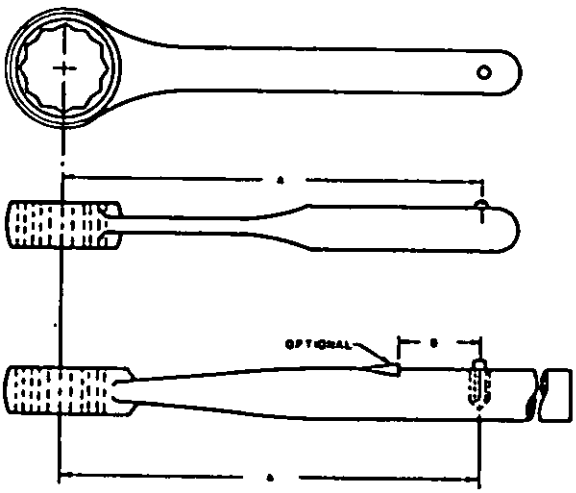


FIGURE 29. Type XVIII, class 1 box wrench, single head, 12-point, stub end, straight.

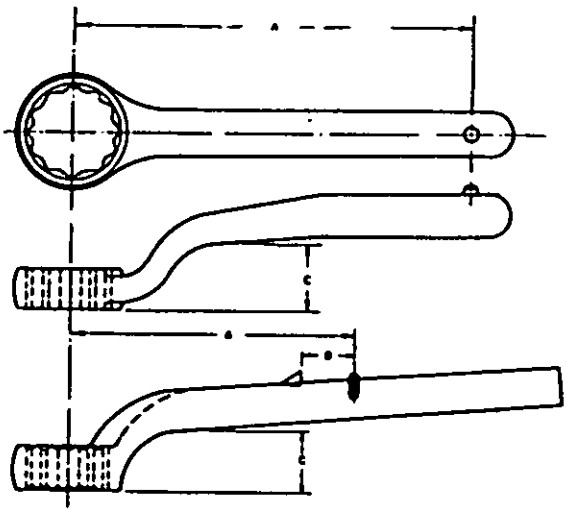


FIGURE 30. Type XVIII, class 2 box wrench, single head, 12-point, stub end, offset.

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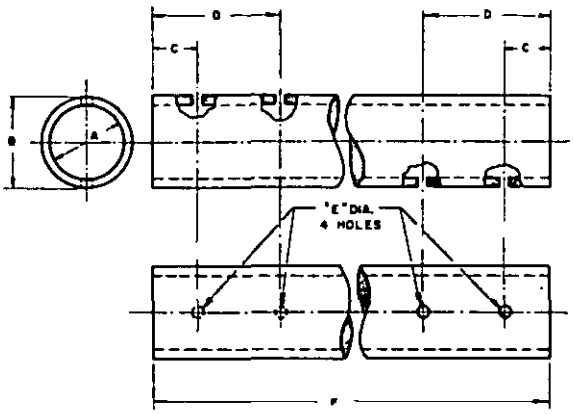


FIGURE 31. Type XIX, handle, extension, (wrench).

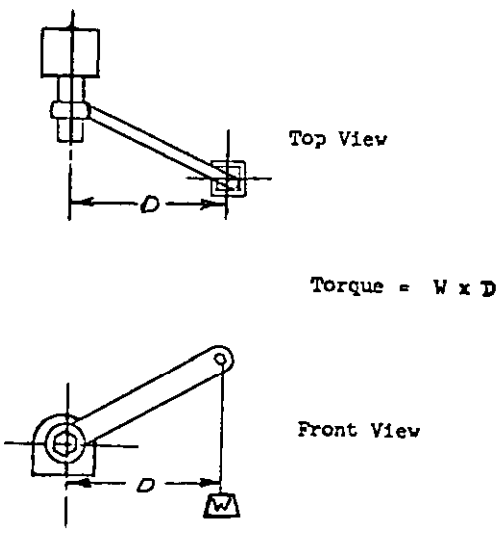


FIGURE 32. Computing for torque load.

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TABLE I. Finish requirements

Type, style, or class	Minimum area of finish
I, II, III, VI, X, XVI, and XVII, classes 2 and 3, also type IV up thru size 1-1/2 x 1-5/8	A minimum of 180° of the outer periphery of the box ends (90° on each side of the longitudinal axis of the wrench) and both faces of the open end shall be bright and shall have a maximum roughness height value of 30 microinches (arithmetical average (A.A.)).
V up thru sizes 2-7/16 and XV	Both faces of open end wrenches shall have a maximum roughness height value of 30 microinches (A.A.).
IV sizes 1-7/16 x 1-13/16 and larger, V sizes 2-9/16 and larger, VII, VIII, XIII, XIV, and XVII Classes 1 and 4	Both faces of each head and the handle extensions (Type XIX) shall have a maximum surface roughness of 125 microinches (A.A.).
IX, XI, XII	Forge flash shall be removed to the extent that it shall blend with adjacent surfaces in keeping with good commercial practice.

TABLE IA. Test loads for box wrenches

Box size	Inch-pounds
5/32	110
3/16	165
13/64	165
7/32	165
15/64	184
1/4	220
9/32	248
5/16	275
11/32	275
3/8	605
7/16	715
1/2	1,020
9/16	1,500
5/8	2,200
11/16	2,640
3/4	2,860
13/16	3,300
7/8	3,630
15/16	4,510
1	5,390
1-1/16	5,940
1-1/8	6,490
1-1/4	7,925
1-5/16	8,800
1-3/8	8,975
1-7/16	9,240
1-1/2	11,495
1-5/8	12,925
1-11/16	15,400

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TABLE II. Type I, class 1, style 1, box wrench, double head, 15° offset each end, regular length

Wrench opening (distance across flats)		Outside diameter of head		Permitted eccentricity of wrench openings to outside diameter of heads	Thickness of heads		Overall length	
A	B	A	B		A	B	C	
Small head	Large head	Small head	Large head		Small head	Large head		
		Max.	Max.	Max.	Max.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches
3/8	7/16	5/8	23/32	0.015	21/64	11/32	6-1/2	8-5/16
7/16	1/2	23/32	51/64	.015	11/32	3/8	7-1/2	8-3/4
7/16	9/16	23/32	29/32	.015	11/32	3/8	7-1/2	8-3/4
1/2	9/16	51/64	29/32	.015	3/8	3/8	7-1/2	9-1/8
9/16	5/8	29/32	31/32	.018	3/8	7/16	8-1/4	9-7/8
5/8	11/16	31/32	1-1/16	.018	7/16	15/32	9-7/16	11
5/8	3/4	31/32	1-1/8	.018	7/16	17/32	9-7/16	11-1/2
11/16	3/4	1-1/16	1-5/32	.018	15/32	17/32	10-1/4	11-3/4
3/4	13/16	1-1/8	1-3/16	.018	17/32	9/16	10-3/4	12-1/2
3/4	7/8	1-5/32	1-5/16	.018	17/32	19/32	10-7/8	13-1/4
13/16	7/8	1-1/4	1-5/16	.020	9/16	19/32	11-3/4	14
7/8	15/16	1-5/16	1-3/8	.020	19/32	19/32	12	14
7/8	1	1-5/16	1-7/16	.020	19/32	5/8	12-1/4	14-1/4
15/16	1	1-3/8	1-1/2	.020	5/8	21/32	13-3/16	15-3/4
15/16	1-1/16	1-7/16	1-5/8	.020	41/64	11/16	13-3/4	16-3/8
1	1-1/16	1-7/16	1-5/8	.020	5/8	11/16	14	16-3/8
1	1-1/8	1-7/16	1-5/8	.020	21/32	3/4	14-1/4	16-3/8
1-1/16	1-1/8	1-19/32	1-11/16	.023	45/64	3/4	15-1/16	18
1-1/16	1-1/4	1-19/32	1-7/8	.023	23/32	13/16	15-7/8	18-1/2
1-1/8	1-5/16	1-5/8	1-29/32	.023	3/4	13/16	17	19
1-1/4	1-5/16	1-7/8	1-61/64	.023	13/16	27/32	17-3/8	19-1/2
1-1/4	1-3/8	1-7/8	2-1/16	.023	13/16	7/8	18-1/4	20-1/2
1-1/4	1-7/16	1-29/32	2-3/16	.023	13/16	15/16	19	21-1/4
1-5/16	1-1/2	1-29/32	2-3/16	.027	53.64	1	20	22
1-7/16	1-1/2	2-1/8	2-3/16	.027	15/16	1	21	23
1-7/16	1-5/8	2-13/64	2-7/16	.027	15/16	1-1/16	22	24
1-1/2	1-11/16	2-3/16	2-7/16	.027	1	1-1/16	22-1/2	24-1/2

TABLE III. Type I, class 1, style 2 box wrench, double offset, 15° offset each end, short length

Wrench opening (distance across flats)		Outside diameter of head		Permitted eccentricity of wrench openings to outside diameter of heads	Thickness of heads		Overall length	
A	B	A	B		A	B	C	
Small head	Large head	Small head	Large head		Small head	Large head		
		Max.	Max.	Max.	Max.	Max.	Min.	Max.
Inch	Inch	Inch	Inches	Inch	Inch	Inch	Inches	Inches
1/4	5/16	7/16	1/2	0.015	1/4	9/32	4	5
5/16	3/8	17/32	5/8	.015	9/32	5/16	4	5
3/8	7/16	5/8	23/32	.015	5/16	11/32	4-1/8	5-1/8
7/16	1/2	23/32	25/32	.015	11/32	23/64	4-1/4	5-1/4
7/16	9/16	11/16	29/32	.015	11/32	3/8	4-1/2	5-1/2
1/2	9/16	25/32	29/32	.015	23/64	13/32	4-5/8	5-5/8
9/16	5/8	29/32	31/32	.018	13/32	15/32	5	6
5/8	11/16	31/32	1-1/32	.018	15/32	15/32	5-1/4	6-1/4
5/8	3/4	31/32	1-1/8	.018	15/32	17/32	5-1/4	6-1/4

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TABLE IV. Type I, class 2, style 1 box wrench, double head, deep offset each end, regular length

Wrench openings (distance across flats)		Outside diameter of heads		Permitted eccentricity of wrench opening to outside diameter	Thickness of heads		See note		Height to where offset blends with handle		Overall length
A	B	A	B		A	B	C	D	E	F	O
Small head	Large head	Small head	Large head		Small head	Large head	Small head	Large head	Small head	Large head	
Max.	Max.	Max.	Max.	Max.	Max.	Max.	Max.	Max.	Min.	Min.	Min.
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches	Inches	Inches
5/16	7/16	5/8	21/32	0.015	5/16	11/32	1 3/16	1 3/16	5/16	5/16	6-13/16
11/32	1/2	11/16	23/32	.015	11/32	23/64	1 3/16	1 3/16	5/16	5/8	7 3/4
3/8	9/16	13/16	25/32	.015	3/8	3/8	1 3/16	1 3/16	5/16	5/8	7 7/8
1/2	5/8	7/8	27/32	.015	3/8	3/8	1 3/16	1 3/16	5/8	5/8	7-31/32
9/16	11/16	29/32	29/32	.018	3/8	13/32	1 3/16	1 3/16	5/8	23/32	8 5/8
5/8	3/4	29/32	1 1/32	.018	13/32	15/32	1 3/16	1 3/16	23/32	23/32	9-1/4
11/16	13/16	29/32	1 1/32	.018	13/32	15/32	1 3/16	1 5/16	23/32	13/16	9 3/4
3/4	7/8	1 1/32	1 1/16	.018	15/32	17/32	1 3/16	1 5/16	23/32	13/16	9 7/8
13/16	15/16	1 1/16	1 1/16	.018	15/32	9/16	1 3/16	1 5/16	13/16	15/16	10 1/8
7/8	1	1 1/16	1 1/8	.018	9/16	9/16	1 5/16	1 11/16	13/16	1	11 1/4
15/16	1 1/16	1 1/8	1 3/16	.020	9/16	5/8	1 11/16	1 11/16	1	1	11 3/8
1	1 1/8	1 1/8	1 3/16	.020	5/8	5/8	1 11/16	2 1/16	1	1 3/32	12
1 1/16	1 3/16	1 3/16	1 3/16	.020	5/8	5/8	1 11/16	2 1/16	1	1 3/32	12 1/8
1 1/8	1 1/4	1 1/4	1 5/16	.020	5/8	11/16	2 1/16	2 1/16	1 3/32	1 3/32	13 1/8
1 3/16	1 5/16	1 5/16	1 5/16	.020	11/16	11/16	2 1/16	2 1/16	1 3/32	1 3/32	14
1 1/4	1 3/8	1 3/8	1 3/8	.020	11/16	11/16	2 1/16	2 1/16	1 3/32	1 3/32	14
1 5/16	1 7/16	1 7/16	1 7/16	.020	11/16	3/4	2 1/16	2 1/16	1 3/32	1 3/32	14 1/8
1 3/8	1 1/2	1 1/2	1 1/2	.023	11/16	3/4	2 1/16	2 1/16	1 1/8	1 1/8	15-13/16
1 7/16	1 9/16	1 9/16	1 9/16	.023	11/16	13/16	2 1/16	2 1/16	1 1/8	1 1/8	16 1/4
1 1/2	1 5/8	1 5/8	1 5/8	.023	3/4	13/16	2 1/16	2 1/16	1 1/8	1 1/8	17
1 9/16	1 11/16	1 5/8	1 5/8	.023	3/4	13/16	2 1/16	2 1/16	1 1/8	1 1/8	17
1 5/8	1 3/4	1 11/16	1 11/16	.023	13/16	13/16	2 1/16	2 1/16	1 1/8	1 1/8	17 1/8
1 11/16	1 13/16	1 13/16	1 13/16	.023	13/16	7/8	2 1/16	2 1/16	1 1/8	1 1/8	18
1 3/4	1 7/8	1 13/16	2 1/16	.023	7/8	13/16	2 1/16	2 1/16	1 5/16	1 5/16	18
1 13/16	1 15/16	1 15/16	2 1/16	.027	7/8	1	2 1/16	2 1/16	1 5/16	1 5/16	10 1/8
1 7/8	1 15/16	2 1/16	2 1/16	.027	7/8	1	2 1/16	2 1/16	1 5/16	1 5/16	10 1/8
1 13/16	1 15/16	2 1/16	2 1/16	.027	15/16	1	2 1/16	2 1/16	1 5/16	1 5/16	20
1 15/16	1 15/16	2 1/16	2 3/16	.027	15/16	1	2 1/16	2 1/16	1 5/16	1 5/16	21 1/8
1 1/4	1 13/16	2 1/16	2 3/16	.027	1	1 1/16	2 1/16	2 1/16	1 5/16	1 5/16	21 1/8

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TABLE V. Type I, class 2, style 2 box wrench, double end, deep offset each end, short length.

Wrench openings (distance across flats)		Outside diameter of heads		Permitted eccentricity of wrench openings to outside diameter of heads	Thickness of heads		See note		Height to where offset blends with handle		Overall length	
A	B	A	B		A	B	C	D	E	F	G	
Small head	Large head	Small head	Large head		Small head	Large head	Small head	Large head	Small head	Large head		
		Max.	Max.	Max.	Max.	Max.	Max.	Max.	Min.	Min.	Min.	Max.
Inch	Inch	Inch	Inches	Inch	Inch	Inch	Inches	Inches	Inch	Inch	Inches	Inches
5/16	13/64	25/64	1 1/32	0.015	5/32	11/64	29/32	29/32	7/16	1/4	2 3/4	3
11/32	15/64	25/64	25/64	.015	3/16	13/64	31/32	1-1/32	1/2	9/32	3	3 1/8
1/4	9/32	29/64	29/64	.015	13/64	7/32	1-1/16	1-1/8	1/2	9/32	3 3/8	4 3/8
1/4	5/16	7/16	1/2	.015	1/4	9/32	1-1/4	1-3/8	25/64	5/16	4	5
5/16	11/32	1/2	9/16	.015	1/4	9/32	7/8	1	15/64	11/32	3 3/4	5
5/16	3/8	9/16	19/32	.015	9/32	5/16	15/16	15/16	15/32	1/2	4	5
3/8	7/16	5/8	23/32	.015	5/16	11/32	1	1	1/2	1/2	4 1/4	5 1/4
7/16	1/2	11/16	25/32	.015	11/32	23/64	1	1 1/16	9/16	9/16	4 1/4	5 1/4
7/16	9/16	11/16	29/32	.015	11/32	7/8	1	1 1/8	9/16	9/16	4 1/4	6
1/2	9/16	13/16	29/32	.015	3/8	7/8	1 1/16	1 1/8	9/16	9/16	5	6
9/16	5/8	25/32	31/32	.018	13/32	15/32	1 1/16	1 1/8	5/8	5/8	5 1/4	6 1/4
5/8	11/16	31/32	1 1/32	.018	13/32	15/32	1 1/8	1 1/8	23/32	23/32	5 1/4	6-25/32
5/8	3/4	31/32	1 1/8	.018	15/32	17/32	1 1/8	1 1/8	13/16	13/16	5 1/4	6 1/4

Note: Center line of socket openings to point where offset blends with handle.

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TABLE VI. Type I, class 2, style 1 box wrench, double head, modified offset regular length

Wrench openings (distance across flats)		Outside diameter of heads		Permitted eccentricity of wrench openings to outside diameter of heads	Thickness of heads		See note		Height to where offset blends with handle		Overall length
A	B	A	B		A	B	C	D	E	F	G
Small head	Large head	Small head	Large head		Small head	Large head	Small head	Large head	Small head	Large head	
		Max.	Max.	Max.	Max.	Max.	Max.	Max.	Min.	Min.	Min.
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches	Inches	Inches
3/8	7/16	19/32	11/16	0.015	5/16	11/32	1-3/16	1-3/16	3/8	3/8	7-1/2
7/16	1/2	11/16	25/32	.015	23/64	23/64	1-3/16	1-7/16	3/8	3/8	7-3/4
7/16	9/16	11/16	29/32	.015	11/32	3/8	1-3/16	1-1/2	3/8	3/8	7-3/4
1/2	9/16	25/32	29/32	.015	3/8	3/8	1-7/16	1-1/2	3/8	3/8	8-1/2
9/16	5/8	29/32	31/32	.018	13/32	15/32	1-1/2	1-9/16	7/16	7/16	8-3/4
5/8	11/16	31/32	1-1/32	.018	15/32	15/32	1-9/16	1-9/16	1/2	1/2	9-3/4
5/8	3/4	31/32	1-1/8	.018	15/32	17/32	1-9/16	1-13/16	1/2	1/2	9-3/4
11/16	3/4	1-1/32	1-1/8	.018	15/32	17/32	1-9/16	1-13/16	1/2	1/2	10
3/4	13/16	1-1/8	1-3/16	.018	17/32	9/16	1-13/16	1-13/16	5/8	5/8	11
3/4	7/8	1-1/8	1-5/16	.018	17/32	19/32	1-13/16	1-13/16	5/8	5/8	11-1/4
13/16	7/8	1-3/16	1-5/16	.020	9/16	19/32	1-13/16	1-13/16	11/16	11/16	11-1/2
7/8	15/16	1-5/16	1-3/8	.020	19/32	19/32	1-13/16	2-9/16	11/16	11/16	12
7/8	1	1-5/16	1-7/16	.020	19/32	5/8	1-13/16	2-9/16	11/16	11/16	12-1/2
15/16	1	1-3/8	1-7/16	.020	39/64	21/32	2-9/16	2-9/16	3/4	3/4	13-1/2
15/16	1-1/16	1-3/8	1-9/16	.020	41/64	11/16	2-9/16	2-9/16	3/4	3/4	14
1	1-1/16	1-7/16	1-9/16	.020	5/8	11/16	2-9/16	2-9/16	13/16	13/16	14-1/4
1	1-1/8	1-15/32	1-5/8	.020	43/64	3/4	2-9/16	2-5/8	13/16	13/16	14-1/2
1-1/16	1-1/8	1-9/16	1-5/8	.023	45/64	3/4	2-9/16	2-5/8	7/8	7/8	14-1/2
1-1/16	1-1/4	1-9/16	1-13/16	.023	23/32	13/16	2-9/16	3-1/8	7/8	7/8	15
1-1/8	1-5/16	1-5/8	1-29/32	.023	3/4	13/16	2-5/8	3-1/8	7/8	7/8	15-1/2
1-3/16	1-5/16	1-5/8	1-29/32	.023	3/4	13/16	2-5/8	3-1/8	7/8	7/8	16
1-1/4	1-5/16	1-13/16	1-29/32	.023	13/16	53/64	3-1/8	3-1/8	7/8	7/8	16
1-1/4	1-3/8	1-13/16	2-1/16	.023	13/16	7/8	3-1/8	3-1/8	1	1	17
1-1/4	1-7/16	1-13/16	2-1/8	.023	13/16	15/16	3-1/8	3-1/8	1	1	18
1-5/16	1-1/2	1-29/32	2-3/16	.027	13/16	1	3-1/8	3-1/8	1	1	19-1/2
1-3/8	1-1/2	2-1/16	2-3/16	.027	13/16	1	3-1/8	3-1/8	1	1	19-1/2
1-7/16	1-1/2	2-1/8	2-3/16	.027	15/16	1	3-1/4	3-1/4	1-1/16	1-1/16	20
1-7/16	1-5/8	2-1/8	2-3/8	.027	15/16	1	3-1/4	3-1/2	1-1/8	1-1/8	21-1/2
1-1/2	1-11/16	2-3/16	2-3/8	.027	1	1-1/16	3-1/4	3-1/2	1-1/8	1-1/8	21-3/4

Note: Centerline of socket openings to point where offset blends with handle.

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TABLE VII. Type I, class 2, style 2 box wrench, double head, modified offset, each end, short length

Wrench opening (distance across flats)		Outside diameter of heads		Permitted eccentricity of wrench openings to outside diameter of heads	Thickness of heads		See note		Height to where offset blends with handle				Overall length	
A	B	A	B		A	B	C	D	E		F		G	
Small head	Large head	Small head	Large head		Small head	Large head	Small head	Large head	Small head		Large head			
		Max.	Max.	Max.	Max.	Max.	Max.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inch	Inch	Inch	Inches	Inch	Inch	Inch	Inches	Inches	Inch	Inch	Inch	Inch	Inches	Inches
3/16	13/64	25/64	1 1/32	0.015	1/8	5/32	3/8	3/8	1/8	9/64	1/8	9/64	2 3/4	3
1/32	15/64	25/64	2 3/64	.015	11/64	3/16	1/8	1/8	3/16	13/64	3/16	13/64	3	3 1/4
1/4	9/32	1 3/32	2 3/64	.015	5/16	13/64	7/16	7/16	5/16	1/4	5/16	1/4	3 1/4	3 7/16
1/4	5/16	7/16	1/2	.015	1/4	9/32	1/2	1/2	7/16	5/16	7/16	5/16	3 1/2	4 1/4
5/16	11/32	1/2	1/2	.015	9/32	9/32	1/2	1/2	5/16	5/16	5/16	5/16	3 3/4	4
5/16	3/8	1/2	15/32	.015	9/32	5/16	9/16	3/4	1/4	15/32	1/4	15/32	4	5
3/8	7/16	19/32	1 1/16	.015	5/16	11/32	5/8	7/8	5/16	15/32	5/16	15/32	4 1/4	5 1/4
7/16	1/2	1 1/16	2 3/32	.015	11/32	23/64	5/8	7/8	5/16	15/32	5/16	15/32	4 1/2	5 3/4
7/16	9/16	1 1/16	2 5/32	.015	11/32	3/8	5/8	7/8	3/8	1/2	3/8	1/2	4 3/4	6
1/2	9/16	2 3/32	2 9/32	.015	23/64	3/8	1 1/16	1 1/16	3/8	1/2	3/8	1/2	5	6
9/16	5/8	2 5/32	2 1/32	.018	13/32	15/32	1 1/8	1 1/8	7/16	5/8	7/16	5/8	5 1/4	6 1/4
5/8	11/16	2 1/32	1 3/32	.018	15/32	15/32	1 1/4	1 5/16	1/2	11/16	1/2	13/16	5 1/2	6 1/2
5/8	3/4	2 1/32	1 1/8	.018	15/32	17/32	1 1/4	1 3/8	1/2	11/16	1/2	11/16	5 3/4	6 3/4

Note: Center line of socket openings to point where offset blend with handle.

TABLE VIII. Type II box wrench, double head, 12-point, half moon

Wrench opening (distance across flats)		Outside diameter of head		Permitted eccentricity of box wrench opening to outside diameter	Thickness of head		Distance between centers of openings (minimum)
Small head	Large head	Small head (maximum)	Large head (maximum)		Small head (maximum)	Large head (maximum)	
Inch	Inch	Inches	Inches	Inch	Inch	Inch	Inches
7/16	1/2	0.695	0.775	0.015	3/8	1/2	4 7/8
9/16	5/8	.868	.946	.018	13/32	7/16	5 13/16
7/8	3/4	.946	1.125	.018	7/16	23/64	6 7/16
1 1/16	3/4	1.025	1.125	.018	23/64	23/64	6 3/4

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TABLE IX. Type III combination wrench, open end, and 15° offset box opening

Wrench opening (distance across flats)	Width of open head (maximum)	Outside diameter of box (maximum)	Permitted eccentricity of box wrench opening to outside diameter (maximum)	Thickness of heads		Overall length (minimum)	Test load
				Open head (maximum)	Box head (maximum)		Open head (minimum)
Inches	Inches	Inches	Inch	Inch	Inch	Inches	Inch-pounds
5/32	7/16	5/16	0.015	9/64	5/32	2-3/4	45
3/16	1/2	11/32	.015	5/32	11/64	2-7/8	45
1/4	5/8	7/16	.015	3/16	1/4	3	67
9/32	5/8	7/16	.015	3/16	1/4	3-9/32	78
5/16	25/32	17/32	.015	13/64	9/32	3-1/4	138
11/32	25/32	17/32	.015	13/64	9/32	3-1/4	193
3/8	7/8	5/8	.015	7/32	21/64	4-3/16	275
7/16	31/32	23/32	.015	1/4	11/32	5	413
1/2	1-5/64	13/16	.015	17/64	23/64	5-1/4	550
9/16	1-15/64	29/32	.018	19/64	3/8	5-3/4	770
5/8	1-3/8	1	.018	21/64	15/32	6-1/8	1,100
11/16	1-1/2	1-3/32	.018	23/64	1/2	7	1,375
3/4	1-5/8	1-5/32	.018	25/64	19/32	8	1,650
13/16	1-53/64	1-11/32	.020	33/64	19/32	10-1/8	2,200
7/8	1-53/64	1-11/32	.020	33/64	19/32	10-1/4	2,475
15/16	2	1-7/16	.020	19/32	11/16	12	3,025
1	2-1/16	1-17/32	.020	19/32	23/32	12-1/2	3,575
1-1/16	2-5/16	1-11/16	.023	5/8	25/32	14	3,850
1-1/8	2-3/8	1-23/32	.023	21/32	27/32	14-1/2	4,400
1-1/4	2-11/16	1-29/32	.023	23/32	29/32	16	5,775
1-5/16	2-45/64	2	.027	23/32	29/32	17-1/2	6,600
1-3/8	2-7/8	2-1/16	.027	3/4	15/16	18	7,425
1-7/16	3	2-3/16	.027	3/4	61/64	18-1/2	8,250
1-1/2	3-1/16	2-3/16	.027	13/16	31/32	19	8,500

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TABLE X. Type IV engineers' wrench, double head, open end, 15°

Wrench openings		Overall length	Width of head		Thickness of head		Test load	
Small head	Large head		Small head	Large head	Small head	Large head	Small head	Large head
		Min.	Max.	Max.	Max.	Max.	Min.	Min.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch-pounds	Inch-pounds
5/32	3/16	2-1/2	7/16	15/32	3/32	3/32	45	45
3/16	1/4	3	15/32	17/32	11/64	11/64	45	67
7/32	1/4	3	1/2	9/16	11/64	11/64	50	67
1/4	5/16	3	19/32	11/16	13/64	13/64	67	138
5/16	11/32	3-5/8	11/16	13/16	13/64	13/64	138	193
5/16	3/8	3-5/8	11/16	13/16	13/64	13/64	138	275
5/16	13/32	3-5/8	11/16	13/16	7/32	7/32	138	325
11/32	3/8	3-5/8	11/16	13/16	13/64	13/64	193	275
3/8	7/16	4-1/2	7/8	31/32	15/64	15/64	275	413
3/8	1/2	4-1/4	7/8	1-5/64	15/64	17/64	275	550
13/32	1/2	4-1/4	13/16	1-5/64	15/64	17/64	325	550
7/16	1/2	5	31/32	1-5/64	17/64	17/64	413	550
7/16	9/16	5	31/32	1-15/64	1/4	19/64	413	770
1/2	9/16	5-1/2	1-5/64	1-15/64	19/64	19/64	550	770
1/2	11/16	6	1-5/64	1-1/2	17/64	11/32	550	1,375
9/16	5/8	6	1-15/64	1-3/8	19/64	21/64	770	1,100
9/16	11/16	6	1-15/64	1-1/2	19/64	11/32	770	1,375
9/16	3/4	6	1-15/64	1-5/8	19/64	3/8	770	1,650
5/8	11/16	7	1-3/8	1-1/2	21/64	11/32	1,100	1,375
5/8	3/4	7	1-3/8	1-5/8	21/64	3/8	1,100	1,650
11/16	3/4	7	1-1/2	1-5/8	11/32	3/8	1,375	1,650
11/16	13/16	7-1/2	1-1/2	1-3/4	11/32	3/8	1,375	2,200
3/4	13/16	8	1-5/8	1-3/4	3/8	3/8	1,650	2,200
3/4	7/8	8	1-5/8	1-7/8	3/8	13/32	1,650	2,475
13/16	7/8	8-1/4	1-3/4	1-7/8	3/8	13/32	2,200	2,475
13/16	15/16	9	1-3/4	2	25/64	27/64	2,200	3,025
7/8	15/16	9	1-7/8	2	13/32	27/64	2,475	3,025
7/8	1	9	1-7/8	2-1/16	27/64	15/32	2,475	3,575
7/8	1-1/16	10	1-7/8	2-1/4	27/64	1/2	2,475	3,850
15/16	1	10	2	2-1/16	27/64	15/32	3,025	3,575
15/16	1-1/16	10	2	2-1/16	27/64	1/2	3,025	3,850
1	1-1/8	10-3/4	2-1/16	2-3/8	15/32	1/2	3,575	4,400
1	1-1/2	11	2-1/16	3-1/16	15/32	5/8	3,575	8,500
1-1/16	1-1/8	11-3/4	2-1/4	2-3/8	1/2	1/2	3,850	4,400
1-1/16	1-1/4	11-3/4	2-1/4	2-9/16	1/2	17/32	3,850	5,775
1-1/8	1-1/4	11-3/4	2-3/8	2-9/16	1/2	17/32	4,400	5,775
1-1/8	1-5/16	12	2-3/8	2-45/64	1/2	9/16	4,400	6,600
1-3/16	1-5/16	13	2-9/16	2-45/64	9/16	9/16	5,000	6,600
1-1/4	1-5/16	13-1/2	2-9/16	2-7/8	17/32	9/16	5,775	6,600
1-1/4	1-3/8	13-1/2	2-9/16	2-7/8	17/32	9/16	5,775	7,425
1-1/4	1-7/16	13-1/2	2-9/16	3	17/32	19/32	5,775	8,250
1-1/4	1-5/8	14	2-9/16	3-1/16	9/16	9/16	5,775	9,000
1-5/16	1-1/2	14	2-45/64	3-1/16	9/16	5/8	6,600	8,500
1-3/8	1-7/16	14	2-7/8	3-1/16	9/16	19/32	7,425	8,250
1-3/8	1-1/2	15-1/2	2-7/8	3-1/16	9/16	5/8	7,425	8,500
1-7/16	1-5/8	15-1/2	3	3-5/16	19/32	41/64	8,250	9,000
1-7/16	1-13/16	15-1/2	3-11/16	4-3/16	15/16	15/16	8,250	14,000
1-1/2	1-5/8	15-1/2	3-1/16	3-5/16	5/8	41/64	8,500	9,000
1-5/8	1-13/16	16-1/2	3-11/16	4-3/16	15/16	15/16	9,000	14,000
1-11/16	1-7/8	18-1/2	4-1/4	4-13/32	1-1/8	1-1/8	12,000	15,000
1-13/16	2	18-1/2	4-1/4	4-13/32	1-1/8	1-1/8	14,000	17,750
2-3/16	2-3/8	21-1/2	5	5-3/4	1-1/4	1-1/4	23,000	28,000
2-9/16	2-3/4	26-1/2	6	6-1/2	1-1/4	1-1/4	35,000	44,000
2-5/8	2-13/16	26-1/2	6	6-1/2	1-1/4	1-1/4	40,000	49,000
2-13/16	3	30	6	7-1/8	1-1/8	1-1/8	45,000	52,000
3-1/8	3-1/2	36-1/8	7-3/16	8-1/16	1-9/16	1-9/16	50,000	55,000

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TABLE XI. Type V engineers' wrench, single head, open end, 15°

Wrench opening (distance across flats)	Overall length (minimum)	Width of head (maximum)	Thickness of head (maximum)	Test load (minimum)
Inches	Inches	Inches	Inches	Inch-pounds
3/8	3 1/16	1	13/64	540
7/16	4 1/16	1 1/16	9/32	700
1/2	4 1/16	1 1/16	9/32	1, 170
9/16	5 1/16	1 13/32	5/16	1, 640
5/8	6 1/16	1 13/16	11/32	2, 600
11/16	6 1/16	1 11/16	11/32	2, 250
3/4	7 1/16	1 13/16	11/32	3, 200
13/16	8	2 1/16	1/2	3, 450
7/8	8	2 1/16	1/2	3, 910
15/16	9 1/16	2 1/16	9/16	4, 860
1	9 1/16	2 1/16	9/16	5, 850
1 1/16	10 1/16	2 13/16	5/8	6, 390
1 1/8	10 1/16	2 13/16	5/8	6, 390
1 1/4	11 1/16	3 1/16	3/4	8, 500
1 5/16	11 1/16	3 1/16	3/4	9, 450
1 3/8	11 1/16	3 1/16	3/4	9, 000
1 7/16	13 1/16	3 1/16	3/4	11, 300
1 1/2	13 1/16	3 1/16	3/4	12, 420
1 5/8	14 1/16	4	7/8	14, 850
1 13/16	14 1/16	4	7/8	17, 250
1 13/16	16 1/16	4 1/16	15/16	22, 500
1 7/8	16 1/16	4 1/16	15/16	25, 920
2	18	4 13/16	1	33, 480
2 1/16	18	4 13/16	1	37, 800
2 3/16	10 1/16	5	1 1/16	40, 620
2 1/4	10 1/16	5	1 1/16	51, 660
2 5/16	21 1/16	6	1 1/16	61, 560
2 7/16	21 1/16	6	1 1/16	66, 600
2 9/16	21 1/16	6	1 1/16	76, 000
2 5/8	23	6	1 1/16	35, 000
2 3/4	25	7	1 13/16	25, 000
2 13/16	26	6 1/16	1 1/16	50, 000
3	28	7 1/16	1 13/16	33, 000
3 1/16	28	7 1/16	1 13/16	35, 000
3 1/8	31	8 1/16	1 13/16	52, 000
3 1/4	33	9 1/16	1 1/16	65, 000
3 3/16	35	9 1/16	1 13/16	68, 000
4 1/16	35	9 1/16	1 13/16	70, 000
4 1/8	40	10 1/16	2 1/16	100, 000

TABLE XII. Type VI, ignition wrench, double head, open end, 15° and 60° heads

Wrench opening (distance across flats)		Width of head (maximum)	Thickness of head (maximum)	Overall length (minimum)	Test (minimum)
15°	60°				
Inch	Inch	Inch	Inch	Inches	Inch-pounds
13/64	13/64	1/4	1/16	3	40
1/4	1/4	1/4	1/16	3	40
13/64	13/64	1/4	1/16	3	40
1/4	1/4	1/4	1/16	3	40
9/32	9/32	5/16	5/32	3 1/16	65
5/16	5/16	5/16	5/32	3 1/16	65
11/32	11/32	5/16	5/32	3 1/16	110
3/8	11/32	5/16	5/32	3 1/16	110
3/8	3/8	1	13/64	4 1/16	250
7/16	7/16	1	13/64	4 1/16	250

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TABLE XIII. Type VII setscrew wrench, double head, open end, 22-1/2°

Wrench openings (distance across flats)		Overall length		Test load (minimum)	
Small head	Large head	Minimum	Maximum	Small head	Large head
Inch	Inch	Inches	Inches	Inch-pounds	Inch-pounds
5/16	7/16	4 3/8	5 5/8	400	700
7/16	1/2	5 3/8	7 3/8	1,000	1,500
1/2	5/8	7 3/8	9 3/8	2,000	3,000
3/4	7/8	10	13	4,000	6,000
1	1 1/8	12 1/4	15 3/4	9,000	12,000

Note: Width across flat of setscrew head is equal to the nominal diameter of the screw.

TABLE XIV. Type VIII setscrew wrench, single head, 22-1/2°, class 1, open end, and class 2, box

Wrench opening (distance across flats)	Overall length		Test load (minimum)
	Minimum	Maximum	
Inches	Inches	Inches	Inch-pounds
1/4	3 1/4	4 1/4	250
5/16	3 3/8	5 1/8	400
3/8	4 3/8	5 5/8	700
7/16	5 1/4	6 3/4	1,000
1/2	5 1/2	7 1/8	1,500
9/16	6 1/2	8 1/2	2,000
5/8	7 1/2	9 1/2	3,000
3/4	8 3/4	11 1/4	4,000
7/8	9 1/8	11 7/8	6,000
1	10 1/4	13 1/2	9,000
1 1/8	10 7/8	14 1/8	12,000

Note: Width across flats of setscrew head is equal to the nominal diameter of the screw.

TABLE XV. Type IX, class 1 structural wrench, offset handle, single head, open end

Wrench opening (distance across flats)	Overall length		Radius R		Test load (minimum)
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inches	Inches	Inch-pounds
7/16	7 1/8	9 1/8	1 1/16	1 5/16	1,200
1/2	7 1/8	9 1/8	1 1/16	1 5/16	1,450
9/16	7 1/2	9 1/4	1 1/16	1 5/16	1,550
5/8	7 1/2	12 1/4	3/4	1	2,150
11/16	9 1/4	12 1/2	3/4	1	2,500
3/4	9 1/4	12 1/8	3/4	1	2,900
13/16	9 1/4	14 3/4	13/16	1 5/16	3,600
7/8	11	14 3/4	7/8	1 1/8	4,300
15/16	11	14 3/4	7/8	1 1/8	4,550
1	11	14 3/4	7/8	1 3/8	5,300
1 1/16	12 3/4	17 3/4	1	1 1/4	6,000
1 1/8	12 3/4	17 3/4	1	1 1/4	6,700
1 1/4	14 3/4	19 1/2	1 1/8	1 3/4	7,700
1 5/16	14 3/4	19 1/2	1 1/8	1 3/8	9,100
1 7/16	16 3/4	21 1/4	1 1/4	1 1/2	10,700
1 1/2	16 3/4	21 1/2	1 1/4	1 1/2	12,200
1 5/8	9 1/4	10 1/4	1 1/4	1 1/2	12,200
1 3/4	18 3/4	23 1/2	1 3/8	1 5/8	13,900
1 15/16	18 3/4	23 1/2	1 3/8	1 5/8	15,600
1 13/16	23	26	1 1/2	1 3/4	17,400
1 7/8	23	26	1 1/2	1 3/4	19,200
2	23	26	1 1/2	1 3/4	21,600

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TABLE XVI. Type IX, class 2 structural wrench, offset handle, box, single head, 12-point

Wrench opening (distance across flats)	Length (approximate)	Heads		Handle offset		Test loads (minimum)
		Thickness (maximum)	Outside diameter (approximate)	At head (approximate)	At end (approximate)	
Inches	Inches	Inches	Inches	Inches	Inches	Inch-pounds
11/16	12	11/16	1 17/32	1 1/4	1 9/32	4,500
3/4	12	11/16	1 19/32	1 3/8	1 23/32	5,250
15/16	12	13/16	1 3/4	1 3/4	1 61/64	6,000
1	12	13/16	1 25/32	1 3/4	1 61/64	7,000
1 1/16	15	7/8	1 61/64	1 7/8	2 5/64	8,000
1 1/8	15	7/8	1 63/64	1 15/16	2 5/64	9,000
1 1/4	17	15/16	2 1/64	1 15/16	2 11/64	10,500
1 5/16	17	15/16	2 3/64	1 15/16	2 11/64	13,000
1 7/16	21	1 1/8	2 5/64	2	2 1/4	15,000
1 1/2	21	1 1/8	2 7/64	2 1/4	2 1/4	17,500
1 5/8	22	1 3/16	2 9/64	2 1/4	2 7/8	20,000
1 11/16	22	1 3/16	2 11/64	2 1/4	2 7/8	23,000
1 13/16	23	1 1/4	2 29/32	2 5/16	3	26,000
1 7/8	23	1 1/4	2 29/32	2 5/16	3	29,000
2	24 1/2	1 5/16	3 3/32	2 5/16	3 5/32	32,000
2 1/16	26	1 5/16	3 5/32	2	3 1/2	35,000
2 1/8	28	1 3/8	3 7/32	2 1/4	3 11/16	40,000
2 5/16	30	1 3/8	3 7/32	2 1/4	3 13/16	45,000
2 1/4	30	1 7/16	4 5/32	2 5/16	3 29/32	50,000

TABLE XVII. Type X, class 1 tappet wrench, double spearhead, open end, 15°

Wrench opening (distance cross flats)		Head width (maximum)		Head thickness (maximum)	Overall length (minimum)	Test loads	
Small head	Large head	Small	Large			Small head	Large head
						Minimum	Minimum
Inch	Inch	Inches	Inches	Inch	Inches	Inch-pounds	Inch-pounds
11/32		1	1 3/8	5/16	7	225	300
3/16	1/4	1 3/16	1 1/4	5/16	7	260	300
7/16	11/32	1 5/16	1 1/4	5/16	7	260	300
1/2	3/16	1 5/16	1 1/4	5/16	7 3/4	300	400
9/16	3/8	1 5/16	1 1/2	5/16	7 3/4	400	600
5/8	13/16	1 5/16	1 1/2	13/64	8	600	800
13/16	1/2	1 3/8	1 3/8	7/32	8	800	900
3/4	15/16	1 3/8	1 13/16	7/32	8	900	1,000
7/8	5/8	1 7/16	1 13/16	7/32	8	900	1,050
13/16	5/8	1 13/16	1 13/16	7/32	8	1,000	1,050

TABLE XVIII. Type X, class 2 tappet wrench, single spearhead, open end, 15°

Wrench opening (distance across flats)	Head width (maximum)	Head thickness (maximum)	Overall length		Test load (minimum)
			minimum	maximum	
Inch	Inches	Inch	Inches	Inches	Inch-pounds
11/32	1 5/32	7/32	11 3/16	15	200
3/8	1 1/4	7/32	11 3/16	15	300
9/16	1 5/16	13/64	11 3/16	15	400
5/8	1 3/8	13/64	11 3/16	15	525
13/16	1 5/16	15/64	11 3/16	15	775
3/4	1 3/16	17/64	11 3/16	15	900
15/16	1 13/16	19/64	11 3/16	15	1,000
1	1 13/16	19/64	11 3/16	15	1,000

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TABLE XIX. Type XI, style 1 slugging wrench, 12-point box, short length

Wrench opening (distance across flats)	Head		Slugging block		Overall length (minimum)
	Outside diameter (minimum)	Thickness (maximum)	Width (minimum)	Thickness (minimum)	
A	B	C	D	E	F
Inches	Inches	Inches	Inches	Inches	Inches
1 1/4	2 5/16	15/16	1 5/16	5/8	6
1 5/16	2 3/8	15/16	1 5/16	5/8	6
1 3/8	2 7/16	1	1 3/8	5/8	6
1 7/16	2 1/2	1	1 7/16	5/8	6
1 1/2	2 7/16	1 1/16	1 1/2	13/16	7
1 9/16	2 1/2	1 1/16	1 1/2	13/16	7
1 13/16	2 11/16	1 1/4	1 5/8	7/8	7
1 7/8	2 3/4	1 1/4	1 5/8	7/8	7
2	3 1/4	1 7/16	2	1	7 1/2
2 1/16	3 1/4	1 1/2	2	1	7 1/2
2 1/8	3 1/2	1 1/2	2	1	7 1/2
2 3/16	3 3/4	1 11/16	2 1/8	1 1/8	8 1/16
2 5/16	4 1/8	1 3/4	2 1/8	1 1/4	8 1/16
2 3/8	4 1/4	1 3/4	2 1/8	1 1/4	8 1/16
2 7/16	4 5/16	1 13/16	2 1/4	1 3/8	9 1/2
2 1/2	4 3/8	2	2 1/4	1 3/8	9 1/2
3	4 3/8	2	2 1/4	1-3/8	9 1/2
3 1/8	4 3/8	2	2 1/4	1-3/8	9 1/2
3 1/4	5 1/4	2 1/16	3 1/8	1-9/16	10
3 1/2	5 1/4	2 1/16	3 1/8	1-9/16	10
3 3/4	5 3/4	2 5/16	3-3/8	1-9/16	10 3/4
3 7/8	5 7/8	2 5/16	3-3/8	1-9/16	10 3/4
4 1/4	6 1/2	2 1/2	3 3/4	1-9/16	11 1/2
4 1/4	6 1/4	2 1/2	3 3/4	1-9/16	11 1/2
4 1/2	6 3/8	2 7/8	3 7/8	1-11/16	11 5/8
4 3/8	6 3/8	2 7/8	3 7/8	1-11/16	11 5/8

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TABLE XX. Type XI, style 2, slugging wrench, 12-point box, long length

Wrench opening (distance across flats)	Head		Sledge block			Offset	Overall length (minimum)
	Outside diameter (maximum)	Thickness (maximum)	Width (maximum)	Thickness (maximum)	Length (maximum)	(Maximum)	
A	B	C	D	E	H	O	F
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1 7/16	2 1/16	1 1/2	1 1/2	1 1/16	2 1/4	3/4	7
1 1/2	2 1/16	1 1/2	1 1/2	1 1/16	2 1/4	3/4	7
1 9/16	2 1/16	1 1/4	1 1/2	1 1/16	2 1/4	1 1/4	8 1/4
1 5/8	3	1 5/8	2	1 1/2	2 3/4	1 1/4	9
1 11/16	3	1 5/8	2	1 1/2	2 3/4	1 1/4	9
1 3/4	3	1 5/8	2	1 1/2	2 3/4	1 1/4	10
1 13/16	3	1 5/8	2	1 1/2	2 3/4	1 1/4	10
1 7/8	3	1 5/8	2	1 1/2	2 3/4	1 1/4	10
2	3 1/4	1 5/8	2 1/4	1 1/16	2 1/2	1 3/8	10 1/4
2 1/16	3 1/4	1 5/8	2 1/4	1 1/16	2 1/2	1 3/8	10 1/4
2 1/8	3 1/4	1 3/4	2 1/4	1 1/2	2 1/2	1 1/2	10 1/4
2 1/4	3 1/4	1 3/4	2 1/4	1 1/2	2 1/2	1 1/2	10 1/4
2 5/16	4 1/4	2	2 1/4	1 1/2	2 1/2	1 1/2	11 1/4
2 3/8	4 1/4	2	2 1/4	1 1/2	2 1/2	1 1/2	11 1/4
2 7/16	4 1/4	2	2 1/4	1 1/2	3	1 1/2	12 1/4
2 1/2	4 1/4	2	2 1/4	1 1/2	3	1 1/2	12 1/4
2 5/8	4 1/4	2 1/4	3	1 1/16	3 1/4	1 1/4	14
2 11/16	4 1/4	2 1/4	3	1 1/16	3 1/4	2	14
2 3/4	4 1/4	2 1/4	3	1 1/16	3 1/4	2	14
3	4 1/4	2 1/4	3 1/4	1 1/16	3 1/4	2	14
3 1/16	4 1/4	2 1/4	3 1/4	1 1/16	3 1/4	2	15 1/4
3 1/8	5 1/4	2 1/4	3 1/4	1 1/4	3 1/2	2 1/4	17 1/4
3 1/4	5 1/4	2 1/4	3 1/4	1 1/4	3 1/2	2 1/4	17 1/4
3 1/8	6	3	3 1/4	1 1/4	3 1/2	2 1/4	17 1/4
3 1/4	6	3	3 1/4	1 1/4	3 1/2	2 1/4	17 1/4
4 1/16	6 1/4	3 1/4	3 1/4	1 1/4	4	2 1/2	18
4 1/4	6 1/4	3 1/4	3 1/4	1 1/4	4	2 1/2	18
4 1/4	7	3 1/4	3 1/4	1 1/4	4	2 1/4	19 1/4
4 1/4	7 1/4	3 1/4	3 1/4	1 1/4	4	2 1/4	19 1/4

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TABLE XXI. Type XII striking wrench, 12-point box

Wrench opening (distance across flats)	Head		Striking block			Diameter of handle (minimum)	Overall length (minimum)	Radius of offset (minimum)
	Outside diameter (minimum)	Thickness (minimum)	Width (minimum)	Thickness (minimum)	Length (minimum)			
A	B	C	D	J	G	E	F	H
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1 7/16	1 17/32	5/16	1	3/4	1 1/2	7/8	8 1/2	1/2
1 1/2	1 17/32	5/16	1	3/4	1 1/2	7/8	8 1/2	1 5/16
1 9/16	1 19/32	5/16	1	7/8	1 3/4	1 1/16	8 1/2	1 5/16
1 5/8	1 19/32	5/16	1	7/8	1 3/4	1 1/16	8 1/2	1 5/16
1 3/4	1 7/8	1 1/16	1 1/8	7/8	1 3/4	3/4	10 1/2	1 1/16
1 13/16	1 7/8	1 1/16	1 1/8	1 1/8	1 3/4	3/4	10 1/2	1 1/8
1 7/8	2 1/8	7/8	1 1/4	1 1/8	1 3/4	7/8	11 1/2	1 1/4
1 15/16	2 3/16	7/8	1 1/4	1 1/8	1 3/4	7/8	11 1/2	1 1/4
1 7/8	2 1 3/32	1	1 1/4	1 1/8	1 3/4	1	12	1 3/8
1 1 1/16	2 1 3/32	1	1 3/4	1 1/4	2	1	12	1 7/16
1 1 3/16	2 1 3/32	1 1/16	1 1/2	1 1/4	2	1	12 1/2	1 3/8
1 1/8	2 3/4	1 1/8	1 1/2	1 1/4	2	1	12 1/2	1 3/8
2	2 1 7/32	1 1/8	1 1/2	1 1/4	2	1	13	1 3/8
2 1/16	3 1/16	1 5/16	1 5/8	1 1/4	2	1	13 1/2	1 7/8
2 1/4	3 1/4	1 5/16	1 5/8	1 1/4	2	1	13 1/2	1 7/8
2 5/16	3 3/8	1 3/8	1 11/16	1 1/4	2 1/2	1 1/8	13 1/2	1 11/16
2 3/8	3 5/8	1 3/8	1 11/16	1 1/4	2 1/2	1 1/8	13 1/2	2 1/4
2 7/16	3 1 1/16	1 3/8	1 13/16	1 1/4	2 1/2	1 1/8	13 1/2	2 1/4
2 1/2	3 7/8	1 3/8	1 11/16	1 1/4	2 1/2	1 1/8	13 1/2	2 1/4
2 1 1/16	4 5/32	1 11/16	1 3/4	1 1/4	2 1/2	1 1/8	15	2 3/8
3	4 1/4	1 3/4	1 3/4	1 1/4	2 1/2	1 1/8	15	2 3/8
3 1/8	4 3/8	1 3/4	1 3/4	1 1/4	2 1/2	1 1/8	15	2 3/8

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TABLE XXII. Type XIII packing-nut wrench, single head, open end, modified 12-point, 30° angle

Wrench opening (distance across flats) A	Circumferential opening (minimum) B	Outside diameter of head (maximum) C	Length of wrench (maximum) D	Thickness of head (maximum) E	Thickness of handle (maximum) F	Test load (minimum)
Inches	Inches	Inches	Inches	Inch	Inch	Inch-pound
¾	0.577	1⅜	6½	1/16	11/64	450
⅞	.671	1⅝	6½	1/16	11/64	550
1	.766	1⅜	6½	1/16	11/64	700
1 1/16	.828	1⅞	6½	1/16	11/64	700
1⅛	.859	1⅞	6½	1/16	11/64	700
1 3/16	.916	2 3/16	7	3/32	5/64	1,300
1¼	1.015	2 3/16	7	3/32	5/64	1,300
1 5/16	1.015	2 3/16	7	3/32	5/64	1,300
1⅜	1.046	2 5/16	7½	3/32	5/64	1,300
1 7/16	1.109	2⅜	7½	3/32	5/64	1,300
1½	1.140	2½	7½	3/32	5/64	1,800
1⅝	1.234	2¾	8	3/32	5/64	1,800
1¾	1.328	2⅞	8	3/32	5/64	1,800

TABLE XXIII. Type XIV flare-nut wrench, single head, modified 12-point

Wrench opening (distance across flats)	Diameter of head (maximum)	Circumferential opening B (minimum)	Thickness of head (maximum)	Length overall (maximum)	Test loads (minimum)
Inches	Inches	Inches	Inches	Inches	Inch-Pounds
7/16	1 3/16	3/32	1/16	6½	---
1/2	1/8	11/32	1 1/32	6½	150
9/16	1 5/16	1/8	1/16	6½	200
5/8	1 13/32	5/16	1/16	6½	275
11/16	1¾	1/4	1 3/32	7½	325
3/4	1 13/16	9/16	1/16	7½	400
13/16	1 13/32	5/16	1/16	7½	500
7/8	1⅞	3/8	1/16	7½	625
15/16	1 15/16	7/16	1/16	7½	800
1	1 17/32	1/2	1/16	7½	1,000
1 1/16	1 17/32	1/2	1/16	7½	1,200
1⅛	1 17/32	1/2	1/16	7½	1,450
1 3/16	1⅞	1/2	1/16	8	1,725
1¼	1 15/16	1	1/16	8	1,900
1 5/16	2 3/32	1 1/16	1/16	10	2,050
1⅜	2 3/16	1⅛	1/16	10	2,200
1 7/16	2 3/32	1⅛	1/16	10½	2,550
1½	2 3/16	1¼	1 1/32	10½	---

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TABLE XXIV. Type XV, thin, double head, open end, 3/16 inch offset, 15° angle wrench

Wrench opening (distance across flats)		A	B	C		D		E		F	G	H	Torque load (minimum)	
				Opening				Radius						
Small head	Large head	Overall length ± 1/8	Radius (maximum)	Small end	Large end	Small end	Large end	Small end	Large end	Width (maximum)	Thickness +.032 −.016	Radius	Small end	Large end
Inches 11/32	Inches 3/8	Inches 5	Inches 13/32	Inches 0.347 +.006 −.000	Inches 0.378 +.006 −.000	Inch 9/32	Inch 3/32	Inch 13/64	Inch 1/32	Inch 3/8	Inch 1/8	Inch 1/8	Inch-pounds 82	Inch-pounds 72
3/8	7/16	5	13/32	.378 +.006 −.000	.440 +.006 −.000	9/32	7/64	1/32	13/64	3/8	1/8	1/8	72	53
7/16	1/2	5 1/2	1/2	.440 +.006 −.000	.504 +.006 −.000	7/64	1/8	17/64	5/16	3/8	1/8	1/8	121	94
1/2	9/16	5 1/2	1/2	.504 +.006 −.000	.566 +.006 −.000	1/8	1/8	5/16	1 1/32	3/8	1/8	1/8	94	72
9/16	5/8	6	1 9/32	.566 +.006 −.000	.629 +.007 −.000	1/8	9/64	1 1/32	1/8	3/8	1/8	1/8	150	120
5/8	11/16	6	1 9/32	.629 +.007 −.000	.692 +.007 −.000	9/64	5/32	1/8	1 3/32	3/8	1/8	1/8	120	125
11/16	3/4	6 1/2	1 11/32	.692 +.007 +.000	.755 +.008 −.000	5/32	11/64	1 3/32	11/64	3/8	1/8	1/8	239	198
3/4	13/16	6 1/2	1 11/32	.755 +.008 −.000	.818 +.008 +.000	11/64	3/16	11/64	1/4	3/8	1/8	1/8	198	158
13/16	7/8	7	1 13/32	.818 +.008 −.000	.880 +.008 −.000	3/16	3/16	1/4	3 5/64	3/8	1/8	1/8	262	219
7/8	15/16	7	1 13/32	.880 +.008 −.000	.944 +.009 −.000	3/16	13/64	3 5/64	3 5/64	3/8	1/8	1/8	219	177
15/16	1	7 1/2	1 15/32	.944 +.009 −.000	1.006 +.009 −.000	13/64	13/64	3 5/64	3 7/64	1/2	1/8	1/8	292	239
1	1 1/16	7 1/2	1 15/32	1.006 +.009 −.000	1.068 +.009 −.000	13/64	13/64	3 7/64	5/16	1/2	1/8	1/8	239	198
1 1/16	1 1/8	8	1 17/32	1.068 +.009 −.000	1.132 +.010 −.000	13/64	13/64	5/16	3 9/32	1/2	3/16	3/16	503	425
1 1/8	1 3/16	8	1 17/32	1.132 +.010 −.000	1.194 +.010 −.000	13/64	7/32	3 9/32	1 1/16	1/2	3/16	3/16	425	356
1 3/16	1 1/4	8 1/2	1 1/2	1.194 +.010 −.000	1.257 +.010 −.000	7/32	7/32	1 1/16	4 3/64	5/8	3/16	3/16	581	500
1 1/4	1 5/16	8 1/2	1 1/2	1.257 +.010 −.000	1.320 +.011 −.000	7/32	9/32	4 3/64	4 5/64	5/8	3/16	3/16	500	425
1 5/16	1 3/8	9	1 1/2	1.320 +.011 −.000	1.383 +.011 −.000	9/32	9/32	4 5/64	4 11/64	5/8	3/16	3/16	669	581
1 3/8	1 7/16	9	1 1/2	1.383 +.011 −.000	1.446 +.011 −.000	9/32	5/16	4 11/64	4 3/64	5/8	3/16	3/16	581	500
1 7/16	1 1/2	9 1/2	1 17/32	1.446 +.012 −.000	1.508 +.012 −.000	5/16	5/16	4 3/64	1/4	3/4	3/16	3/16	762	666
1 1/2	1 9/16	9 1/2	1 17/32	1.508 +.012 −.000	1.571 +.012 −.000	5/16	5/16	1/4	2 5/32	3/4	3/16	3/16	666	581
1 9/16	1 5/8	10	1 23/64	1.571 +.012 −.000	1.634 +.012 −.000	5/16	11/32	2 5/32	1 5/16	3/4	5/16	5/16	912	806
1 5/8	1 11/16	10	1 23/64	1.634 +.012 −.000	1.696 +.012 −.000	11/32	11/32	1 5/16	2 1/32	3/4	3/16	3/16	806	681

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TABLE XXIV. Type XV, thin, double head, open end, 3/16 inch offset, 15° angle wrench (cont.)

Wrench opening (distance across flats)		A	B	C		D		E		F	G	H	Torque load (minimum)	
		Overall length	Radius (maximum)	Opening				Radius		Width (maximum)	Thickness +.032 −.016	Radius		
Small head	Large head	± 1/64		Small end	Large end	Small end	Large end	Small end	Large end				Small end	Large end
1 1/16	1 1/32	10 1/2	1 3/16	1.696 +.012 −.000	1.759 +.013 −.000	1 1/32	1/2	3/32	1	1/2	5/16	5/16	1,072	981
1 1/32	1 3/16	10 1/2	1 3/16	1.759 +.013 −.000	1.822 +.013 −.000	1/2	1/2	1	1 1/64	1/2	5/16	5/16	981	856
1 3/16	1 1/32	11	1 1/32	1.822 +.013 −.000	1.885 +.013 −.000	1/2	1/2	1 1/64	1 1/32	1/2	5/16	5/16	1,100	1,072
1 1/32	1 3/16	11	1 1/32	1.885 +.013 −.000	1.948 +.013 −.000	1/2	1 3/32	1 1/32	1 1/16	1/2	5/16	5/16	1,072	959
1 3/16	2	11 1/2	1 5/16	1.948 +.013 −.000	2.011 +.014 −.000	1 3/32	5/16	1 1/16	1 5/16	1/2	5/16	5/16	1,312	1,184
2	2 5/16	11 1/2	1 5/16	2.011 +.014 −.000	2.074 +.014 −.000	5/16	5/16	1 5/16	1 1/32	1/2	5/16	5/16	1,184	1,069
2 5/16	2 1/16	12	1 3/16	2.074 +.014 −.000	2.136 +.014 −.000	5/16	5/16	1 1/32	1 1/16	5/16	1/4	1/4	2,100	1,921
2 1/16	2 5/16	12	1 3/16	2.136 +.014 −.000	2.200 +.015 −.000	5/16	1/2	1 1/16	1 3/32	5/16	1/4	1/4	1,921	1,737
2 5/16	2 1/16	12 1/2	1 5/32	2.200 +.015 −.000	2.262 +.015 −.000	1/2	1/2	1 3/32	1 5/16	5/16	1/4	1/4	2,679	2,475
2 1/16	2 5/16	12 1/2	1 5/32	2.262 +.015 −.000	2.325 +.016 −.000	1/2	5/16	1 5/16	1 1/32	5/16	1/4	1/4	2,475	1,872
2 5/16	2 1/16	13	2	2.325 +.016 −.000	2.388 +.016 −.000	5/16	5/16	1 1/32	1 1/16	1	1/4	1/4	2,900	2,679
2 1/16	2 5/16	13	2	2.388 +.016 −.000	2.450 +.016 −.000	5/16	5/16	1 1/16	1 1/32	1	1/4	1/4	2,679	2,475
2 5/16	2 1/16	13 1/2	2 1/32	2.450 +.016 −.000	2.514 +.017 −.000	5/16	5/16	1 1/32	1 5/16	1	1/4	1/4	3,117	2,892

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TABLE XXIV. Type XV, thin, double head, open end, 3/16 inch offset, 15° angle wrench (cont.)

Wrench opening (distance across flats)		A	B	C		D		E		F	G	H	Torque load (minimum)	
				Opening				Radius						
Small head	Large head	Overall length ± 1/16	Radius (maximum)	Small end	Large end	Small end	Large end	Small end	Large end	Width (maximum)	Thickness +.032 −.016	Radius	Small end	Large end
2 1/2	2 5/16	13 1/2	2 3/32	2.514 +.017 −.000	2.576 +.017 −.000	5/16	5/16	1 7/16	1 1/2	1	1/4	1/4	2,892	2,679
2 5/16	2 3/8	14	2 5/16	2.576 +.017 −.000	2.639 +.017 −.000	5/16	5/16	1 1/2	1 17/32	1 3/8	1/4	1/4	3,346	3,108
2 3/8	2 7/16	14	2 5/16	2.639 +.017 −.000	2.701 +.017 −.000	5/16	5/16	1 17/32	1 9/16	1 3/8	1/4	1/4	3,108	2,904
2 7/16	2 1/2	14 1/2	2 1/4	2.701 +.017 −.000	2.766 +.017 −.000	5/16	5/16	1 9/16	1 19/32	1 3/8	5/16	5/16	5,031	4,656
2 1/2	2 9/16	14 1/2	2 1/4	2.776 +.017 −.000	2.827 +.018 −.000	5/16	5/16	1 19/32	1 5/8	1 3/8	5/16	5/16	4,656	4,325
2 9/16	2 5/8	15	2 13/32	2.827 +.018 −.000	2.899 +.018 −.000	5/16	11/16	1 5/8	1 21/32	1 1/2	5/16	5/16	6,100	6,656
2 5/8	2 11/16	15	2 13/32	2.890 +.018 −.000	2.954 +.019 −.000	11/16	11/16	1 23/32	1 11/16	1 5/8	5/16	5/16	5,656	5,350
2 11/16	3	15 1/2	2 1/2	2.954 +.019 −.000	3.016 +.019 −.000	11/16	11/16	1 11/16	1 3/4	1 3/4	5/16	5/16	6,488	6,100
3	3 1/16	15 1/2	2 1/2	3.016 +.019 −.000	3.146 +.019 −.000	11/16	13/16	1 3/4	1 13/16	1 1/4	5/16	5/16	6,100	5,537
3 1/16	3 1/4	16	2 11/16	3.146 +.019 −.000	3.268 +.020 −.000	13/16	5/16	1 13/16	1 7/8	1 1/4	5/16	5/16	7,712	6,775
3 1/4	3 5/16	16	2 13/16	3.268 +.020 −.000	3.393 +.021 −.000	5/16	5/16	1 7/8	1 31/32	1 1/4	5/16	5/16	6,775	6,062
3 5/16	3 1/2	16 1/2	2 15/16	3.393 +.021 −.000	3.518 +.021 −.000	5/16	13/16	1 31/32	2 1/32	1 3/8	5/16	5/16	9,531	8,631
3 1/2	3 5/8	16 1/2	2 15/16	3.518 +.021 −.000	3.645 +.021 −.000	13/16	13/16	2 1/32	2 1/8	1 3/8	5/16	5/16	8,631	7,712

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TABLE XXV. Type XVI, style 1 angle wrench, double head, open end, 15° and 60° heads

Wrench opening (distance across flats)	Head width (maximum)	Head thickness (maximum)	Overall length (minimum)	Test load (minimum)
inches	inches	inch	inches	inch-pounds
11/32	25/32	11/64	4 1/2	40
5/16	25/32	1 3/64	4 1/2	40
11/32	25/32	1/16	5	75
13/32	1 1/16	1 3/64	5	250
3/8	1 1/16	1/16	5	100
7/16	1 1/16	1 3/64	6	150
15/32	1 1/16	1 3/64	5 1/2	200
1 1/32	1 1/16	1 5/64	6 1/2	300
1/2	1 13/64	2 1/64	7	500
1 3/32	1 29/64	1 5/64	7 1/2	700
17/32	1 15/64	1 9/64	7 1/2	825
9/16	1 15/64	1 5/64	8 1/2	900
1	2 1/64	2 1/64	9 1/2	1, 700
1 1/16	2 1/64	2 1/64	9 1/2	1, 700
1 1/8	2 13/64	2 1/64	10 1/2	2, 250
1 1/2	2 1/2	2 1/64	11	2, 500
1 5/16	2 17/32	1 1/2	10 1/2	2, 800
1 3/8	2 1/2	2 1/64	11 1/2	3, 200
1 7/16	2 17/32	2 9/64	12	3, 500
1 1/2	3	2 9/64	12 1/2	3, 800

TABLE XXVI. Type XVI, style 2 angle wrench, double head, open end, 30° and 60° heads

Wrench openings (distance across flats)	Head width (maximum)	Head thickness (maximum)	Overall length (minimum)	Test load (minimum)
inch	inches	inch	inches	inch-pounds
3/8	1 1/16	11/32	5	250
7/16	1 1/16	1/4	5 1/2	375
1/2	1 5/32	1/4	6	500
9/16	1 1/4	5/16	6 1/2	700
5/8	1 13/32	5/16	6 1/2	1, 000
11/16	1 13/32	3/8	7 1/2	1, 250
3/4	1 5/16	13/32	7 1/2	1, 500

TABLE XXVII. Type XVII, class 1 thin head wrench, double head, open end 15° (jam nut)

Wrench openings (distance across flats)		Width of heads		Thickness of heads (maximum)	Overall length (minimum)	Test loads	
Small head	Large head	Small head (maximum)	Large head (maximum)			Small head (minimum)	Large head (minimum)
inches	inches	inches	inches	inch	inches	inch-pounds	inch-pounds
11/32	1/2	7/16	1 1/16	5/32	4 1/2	200	250
1/2	9/16	1 1/64	1 1/16	5/32	5 1/2	250	300
1 1/16	1 1/4	2 1/4	2 3/8	5/32	10 1/2	1, 800	1, 950

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TABLE XXVIII. Type XVII, class 2 thin head wrench, single head, open end, straight

Wrench openings (distance across flats)	Width of head (maximum)	Thickness of head (maximum)	Overall length (minimum)	Test load (minimum)
Inches	Inches	Inch	Inches	Inch-pounds
1	2 1/32	1/4	12 1/4	1, 750
1 1/16	2 1/32	1/4	12 1/4	1, 750
1 1/8	2 1/32	1/4	12 1/4	1, 750
1 3/16	2 1/32	1/4	12 1/4	1, 750
1 1/4	2 1/32	1/4	12 1/4	1, 750
1 3/8	2 1/32	1/4	12 1/4	1, 750
1 7/16	2 3/32	9/32	12 1/4	2, 000
1 1/2	2 3/32	9/32	12 1/4	2, 000
1 9/16	2 3/32	9/32	12 1/4	2, 000
1 5/8	2 3/32	9/32	12 1/4	2, 000
1 11/16	2 3/32	9/32	12 1/4	2, 500
1 3/4	3 1/32	5/16	12 1/4	2, 500
1 13/16	3 1/32	5/16	12 1/4	2, 500
1 7/8	3 1/32	5/16	12 1/4	2, 500
2	3 1/32	5/16	12 1/4	2, 500
2 1/16	3 1/32	5/16	12 1/4	2, 500
2 1/8	3 1/32	5/16	12 1/4	2, 500
2 3/16	3 1/32	5/16	12 1/4	2, 500
2 1/4	3 1/32	5/16	12 1/4	2, 500

TABLE XXIX. Type XVII, class 3 thin head wrench, single head, open end, 30° head

Wrench opening (distance across flats)	Width of head (maximum)	Thickness of head (maximum)	Overall length (minimum)	Test load (minimum)
Inches	Inches	Inch	Inches	Inch-pounds
3/4	1 11/16	1/4	6 1/4	1, 000
13/16	1 11/16	1/4	6 1/4	1, 000
7/8	1 11/16	1/4	6 1/4	1, 000
15/16	1 7/8	1/4	6 1/4	1, 000
1	1 7/8	1/4	6 1/4	1, 000
1 1/32	1 11/16	1/4	6 1/4	1, 000
1 1/16	1 7/8	1/4	6 1/4	1, 000
1 1/8	2 1/16	1/4	6 1/4	1, 000
1 3/16	2 1/16	1/4	6 1/4	1, 000
1 1/4	2 1/16	1/4	6 1/4	1, 000
1 5/32	2 1/16	1/4	6 1/4	1, 000
1 3/16	2 1/2	9/32	7 1/4	1, 500
1 3/8	2 1/2	9/32	7 1/4	1, 600
1 7/16	2 1/2	9/32	7 1/4	1, 500
1 1/2	2 1/2	9/32	7 1/4	1, 500
1 9/16	2 1/2	9/32	7 1/4	1, 500
1 5/8	2 1/2	9/32	7 1/4	1, 500
1 11/16	2 1/2	9/32	7 1/4	1, 500
1 3/4	3 1/2	5/16	8 1/2	2, 400
1 13/16	3 1/2	5/16	8 1/2	2, 400
1 7/8	3 1/2	5/16	8 1/2	2, 400
1 15/16	3 1/2	5/16	8 1/2	2, 400
2	3 1/2	5/16	8 1/2	2, 400
2 1/16	3 1/2	5/16	8 1/2	2, 400
2 1/8	3 1/2	5/16	8 1/2	2, 400
2 3/16	3 1/2	5/16	8 1/2	2, 400
2 1/4	3 1/2	5/16	8 1/2	2, 400
2 5/16	3 1/2	5/16	8 1/2	2, 400
2 3/8	3 1/2	5/16	8 1/2	2, 400
2 7/16	3 1/2	5/16	8 1/2	2, 400

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TABLE XXX. Type XVII, class 4 thin head wrench, single head, open end, 15° head

Wrench opening (distance across flats)	Width of heads (maximum)	Thickness of heads (maximum)	Overall length (minimum)	Test load (minimum)
Inches	Inches	Inch	Inches	Inch-pounds
5/16	1 5/16	1/4	3 1/2	320
5/16	1 5/16	1/4	3 1/2	320
3/8	1 3/8	5/16	4 3/16	475
3/8	1 3/8	1/4	5 1/16	650
7/16	1 3/8	1/4	5 1/16	650
1/2	1 29/32	1/4	6 1/2	950
7/16	2 5/32	1/4	6 1/2	1,350
1/2	2 5/32	1/4	6 1/2	1,350
9/16	2 5/16	1/4	7 1/2	1,800
1	2 5/16	1/4	7 1/2	1,800
1 1/16	2 1/2	1/2	8 1/2	2,550
1 1/16	2 1/2	1/2	8 1/2	2,550
1 1/2	3	1/2	9 1/2	3,400
1 5/16	3	1/2	9 1/2	3,780
1 5/16	3 1/8	1/2	11 1/2	5,800
1 1/16	3 1/8	1/2	11 1/2	5,800
1 1/16	3 1/16	1/2	13	9,000
1 1/16	3 1/16	1/2	13	9,000

TABLE XXXI. Type XVIII, class 1 box wrench, single head, 12-point, stub end, straight

Wrench Opening (distance across flats)	Outside diameter of head	Thickness of head	Wrench opening to plunger (centerline to centerline)		Centerline of plunger to stop (stop optional)	Diameter of handle	Overall length	Test load
	Maximum	Maximum	A MIN.	A MAX.	B±1/32	+.000 -.062	Minimum	Minimum
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch-lbs.
1	1-5/8	3/4	4-15/16	7-5/16	1-1/4	7/8	7-13/16	6,150
1-1/16	1-43/64	3/4	4-15/16	7-5/16	1-1/4	7/8	7-13/16	6,500
1-1/8	1-3/4	3/4	4-15/16	7-13/16	1-1/4	7/8	7-7/8	7,000
1-1/4	1-15/16	7/8	4-15/16	8-1/16	1-1/4	7/8	7-15/16	7,250
1-5/16	2-1/16	7/8	6-15/16	8-5/16	1-1/4	7/8	10	8,000
1-3/8	2-3/32	7/8	6-15/16	8-9/16	1-1/4	7/8	10-1/2	9,000
1-7/16	2-11/64	15/16	6-15/16	8-13/16	1-1/4	7/8	10-13/16	9,550
1-1/2	2-5/16	1	6-15/16	9-1/16	1-1/4	7/8	11-1/16	10,450
1-9/16	2-23/64	1	6-15/16	9-9/16	1-1/4	7/8	11-1/2	11,550
1-5/8	2-1/2	1	6-15/16	10-1/16	1-1/4	1	12-1/8	11,750
1-11/16	2-21/32	1-1/16	6-15/16	10-3/16	1-1/4	1	12-3/8	14,000
1-3/4	2-11/16	1-3/32	6-15/16	10-5/16	1-1/4	1	12-9/16	18,000
1-13/16	2-3/4	1-1/8	6-15/16	10-9/16	1-1/4	1	12-13/16	19,000
1-7/8	2-13/16	1-5/32	6-15/16	10-13/16	1-1/4	1	13-1/8	21,800
2	3-1/16	1-5/16	6-15/16	10-13/16	1-1/4	1	13-3/8	29,000
2-3/16	3-7/32	1-5/16	6-15/16	11-1/16	1-1/4	1	13-1/2	29,000
2-1/4	3-13/32	1-11/32	10-15/16	11-9/16	1-1/4	1	13-9/16	29,000
2-3/8	3-7/8	1-7/16	10-9/16	11-9/16	5-1/4	1-1/8	13-11/16	32,000
2-9/16	4-3/16	1-1/2	10-9/16	11-9/16	5-1/4	1-1/8	13-13/16	35,000
2-5/8	4-3/16	1-1/2	10-9/16	11-9/16	5-1/4	1-1/8	13-13/16	35,000
2-3/4	4-1/2	1-19/32	10-9/16	11-9/16	5-1/4	1-1/8	13-15/16	40,000
2-15/16	4-9/16	1-11/16	10-9/16	11-9/16	5-1/4	1-1/8	14-1/16	45,000
3-1/8	4-7/8	1-25/32	10-9/16	11-11/16	5-1/4	1-1/8	14-3/16	48,000

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TABLE XXXII. Type XVIII, class 2 box wrench, single head, 12-point, stub end, offset

Wrench openings (distance across flats)	Outside diameter of head	Thickness of head	Wrench opening to plunger (centerline to centerline) A		Centerline of plunger to stop (stop optional)	Height of offset	Diameter of handle	Overall length	Test load
	Maximum	Maximum	Min.	Max.	B ± 1/32	C-minimum	+0.000 −0.062	Minimum	Minimum
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch-Lbs.
1	1-5/8	3/4	6-5/8	6-3/4	----	15/16	7/8	7-1/2	6,150
1-1/16	1-43/64	3/4	4-15/16	6-13/16	1-1/4	15/16	7/8	7-13/16	6,500
1-1/8	1-3/4	3/4	4-15/16	7-1/8	1-1/4	1	7/8	7-7/8	7,000
1-1/4	1-15/16	7/8	4-15/16	7-11/16	1-1/4	1-1/16	7/8	7-15/16	7,250
1-5/16	2	7/8	6-7/16	8	1-1/4	1-1/8	7/8	9-15/16	8,000
1-3/8	2-3/32	7/8	6-7/16	8	1-1/4	1-3/16	7/8	9-15/16	9,000
1-7/16	2-7/32	15/16	6-7/16	8-3/8	1-1/4	1-1/4	7/8	10-3/8	9,550
1-1/2	2-5/16	15/16	6-7/16	9	1-1/4	1-5/16	7/8	10-5/8	10,450
1-9/16	2-11/32	15/16	6-7/16	9-1/16	1-1/4	1-5/16	7/8	10-5/8	11,550
1-5/8	2-1/2	1	6-7/16	9-1/8	1-1/4	1-3/8	1	10-11/16	11,750
1-11/16	2-5/8	1-1/16	6-15/16	9-9/16	1-1/4	1-7/16	1	11-13/16	14,000
1-3/4	2-11/16	1-1/16	6-15/16	9-9/16	1-1/4	1-1/2	1	11-13/16	18,000
1-13/16	2-3/4	1-5/32	6-15/16	10-1/8	1-1/4	1-9/16	1	12-3/16	19,000
1-7/8	2-13/16	1-1/8	6-15/16	10-3/16	1-1/4	1-5/8	1	12-1/2	21,800
2	3	1-1/4	6-15/16	10-5/16	1-1/4	1-3/4	1	12-3/4	29,000
2-3/16	3-7/32	1-5/16	6-15/16	10-5/16	1-1/4	1-7/8	1	12-13/16	29,000
2-1/4	3-3/8	1-11/32	6-15/16	10-7/16	1-1/4	1-7/8	1	13	29,000
2-3/8	3-7/8	1-3/8	10-7/16	11-9/16	5-1/4	2	1-1/8	13-3/8	32,000
2-9/16	4-3/16	1-9/16	10-7/16	11-9/16	5-1/4	2-1/4	1-1/8	13-3/4	35,000
2-5/8	4-3/16	1-9/16	10-7/16	11-9/16	5-1/4	2-1/4	1-1/8	13-9/16	35,000
2-3/4	4-1/2	1-11/16	9-11/16	11-9/16	5-1/4	2-3/8	1-1/8	12-15/16	40,000
2-15/16	4-9/16	1-3/4	9-11/16	11-9/16	5-1/4	2-3/8	1-1/8	13-1/8	45,000
3-1/8	4-7/8	1-3/4	9-11/16	11-9/16	5-1/4	2-3/8	1-1/8	13-1/8	48,000

TABLE XXXIII. Type XIX, handle, extension (wrench)

Size A	For wrench size	Handle dimensions B	C	D	E	F
+0.015 −0.000		Maximum	+0.000 −0.0625	+0.000 −0.0625	+0.010 −0.000	± 1/4
Inches 7/8	Inches 1 through 1 5/8	Inches 1 3/4	Inches 1 1/4	Inches 3	Inch 1/2	Inches 24
1	1 13/16 through 2 1/4	1 13/16	1 1/4	3 1/8	5/16	30
1 1/8	2 5/8 through 3 1/8	1 1/2	----	5 1/4	5/16	36

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TABLE XXXIV. Sampling data

Category	Sample unit	Inspection level	AQL for major defectives	AQL for minor defectives	AQL expressed in terms of
Visual	One component each	S-4	4.0	6.5	Percent defective
Dimensional and testing	One component each	S-3	4.0	6.5	Percent defective

TABLE XXXV. Classification of defects

Defects	Classification	
	Major	Minor
Type, class, style, form, or size not as specified.	X	
Material, coating, marking not as specified.	X	
Roughness height value 20 percent greater than specified.	X	
Roughness height value greater than specified by less than 20 percent.		X
Dimensions for wrench openings out of tolerance.	X	
Other dimensions out of tolerance by 5 percent or more of allowable dimension.	X	
Other dimensions out of tolerance by less than 5 percent of allowable dimension.		X
Discontinuity in coating.		X
Item marking readable but not clear.		X
Any failure to conform to specified requirements that may affect usability.	X	
Any failure to conform to specified requirements that is not likely to affect usability.		X

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TABLE XXXVI. Hexagon mandrel dimensions (inches)

Nominal size of wrench opening	Hexagon mandrel dimensions	
	Across flats tolerances	Across corners minimum
5/32	+.001 -.002	.1745
3/16	+.001 -.002	.2095
7/32	+.001 -.002	.2440
1/4	+.001 -.002	.2780
9/32	+.001 -.002	.3133
5/16	+.001 -.002	.3495
11/32	+.001 -.002	.3860
3/8	+.001 -.002	.4225
7/16	+.001 -.002	.4935
1/2	+.001 -.003	.5635
9/16	+.001 -.003	.6339
5/8	+.001 -.003	.7055
11/16	+.001 -.003	.7769
3/4	+.001 -.003	.8485
13/16	+.001 -.003	.9201
7/8	+.001 -.003	.9917
15/16	+.001 -.003	1.0631
1	+.001 -.003	1.1297
1-1/16	+.001 -.003	1.2013
1-1/8	+.001 -.003	1.2728
1-3/16	+.001 -.003	1.3443
1-1/4	+.001 -.003	1.416
1-5/16	+.001 -.003	1.487
1-3/8	+.001 -.003	1.559
1-7/16	+.001 -.003	1.631
1-1/2	+.001 -.003	1.702
1-9/16	+.001 -.007	1.770
1-5/8	+.001 -.007	1.841
1-11/16	+.001 -.007	1.912
1-3/4	+.001 -.007	1.983
1-13/16	+.001 -.007	2.054
1-7/8	+.001 -.007	2.124
1-15/16	+.001 -.007	2.195
2	+.001 -.007	2.266
2-1/16	+.001 -.007	2.337
2-1/8	+.001 -.007	2.408
2-3/16	+.001 -.007	2.479
2-1/4	+.001 -.007	2.549
2-5/16	+.001 -.007	2.621
2-3/8	+.001 -.007	2.691
2-7/16	+.001 -.007	2.762
2-1/2	+.001 -.007	2.833
2-9/16	+.001 -.008	2.903
2-5/8	+.001 -.008	2.974
2-3/4	+.001 -.008	3.116
2-13/16	+.001 -.008	3.187
2-15/16	+.001 -.008	3.328
3	+.001 -.008	3.399
3-1/8	+.001 -.008	3.541
3-1/4	+.001 -.010	3.682
3-3/8	+.001 -.010	3.824
3-1/2	+.001 -.010	3.966
3-3/4	+.001 -.010	4.249
3-7/8	+.001 -.010	4.391
4	+.001 -.010	4.532
4-1/8	+.001 -.010	4.674
4-1/4	+.001 -.010	4.816
4-1/2	+.001 -.010	5.099
4-5/8	+.001 -.012	5.240
4-7/8	+.001 -.012	5.524
5	+.001 -.012	5.665

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TABLE XXXVII. Square mandrel dimensions and maximum depth of mandrel insertion (inches)

Size across flats	Across flats tolerance	Across corners minimum
3/16	+.001 -.003	0.2577
7/32	+.001 -.003	.3036
1/4	+.001 -.003	.3436
5/16	+.001 -.003	.4294
3/8	+.001 -.003	.5153
7/16	+.001 -.003	.6012
1/2	+.001 -.003	.687
9/16	+.001 -.003	.773
5/8	+.001 -.003	.859
11/16	+.001 -.003	.945
3/4	+.001 -.003	1.031
13/16	+.001 -.003	1.117
7/8	+.001 -.003	1.202
15/16	+.001 -.003	1.288
1	+.001 -.003	1.374

TABLE XXXVIII. Type XI slugging wrench, box, 12-point

Styles	Size of wrenches	Magnitude of impact blow	
		Weight pounds	Height dropped feet
1—Short length 2—Long length	All sizes	24	15
	1 3/16 inch through 1 1/16 inch	10	10
	2 inch through 2 1/16 inch	15	10
	2 3/16 inch through 4 1/16 inch	24	15

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity

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U. S. GOVERNMENT PRINTING OFFICE : 1975 - 585-649/1503

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